Lecture 4

Clustering, Probability Review

An introduction to unsupervised learning
and a review of core concepts in probability

EECS 189/289, Spring 2026 @ UC Berkeley

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Roadmap

- Supervised vs Unsupervised Learning
- Clustering
  - K-Means clustering
- Probability (Review)
  - Joint Distributions
  - The IID assumption.

Roadmap

- Supervised vs Unsupervised Learning
- Clustering
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First, recall supervised learning

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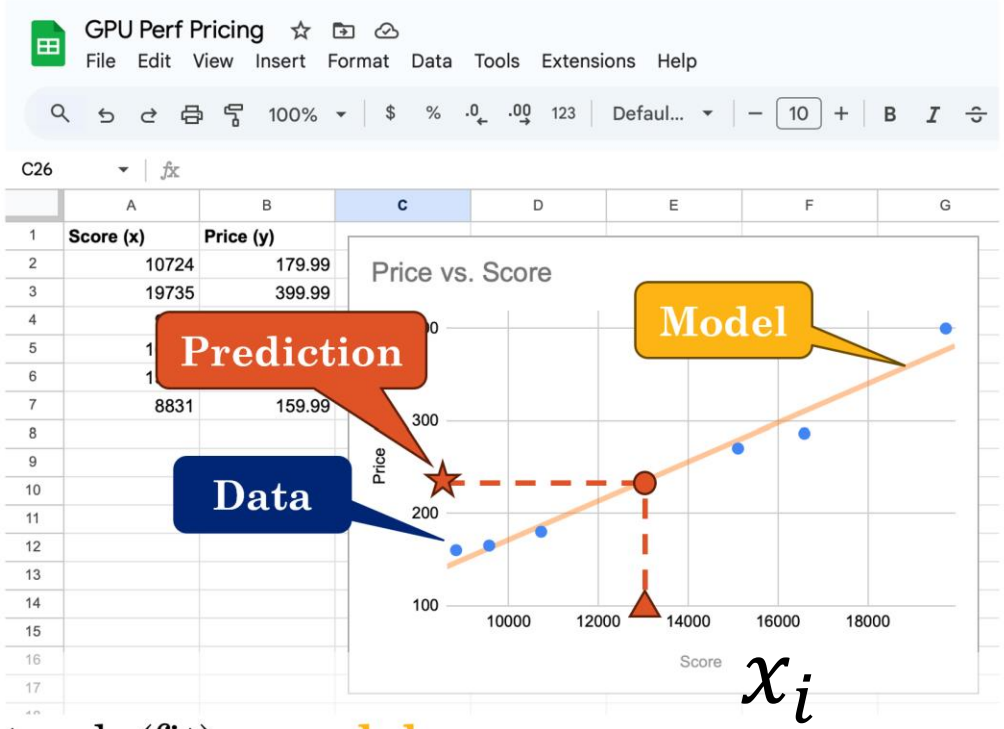
Supervised learning, $\mathrm{D} = \{(x_i, y_i)\}$ for $x \in \mathbb{R}^d$ and $y \in \mathbb{R}$ or $y \in \mathbb{Z}$.

e.g., **predict** house price, y_i, from zip code and square footage, x_i

Linear Regression is Machine Learning



y_i

x_i

Basic Recipe:

Training: Use **data** to teach (fit) a **model**.

Inference: Use the **model** to make **predictions** (decisions).

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Unsupervised learning

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Much ML is focused on modeling $\mathbf{D} = \{x_i\}$, *unsupervised learning*

An eruption in Old Faithful geyser, USA

$x_i = (x_i^1, x_i^2)$

x^1 Duration of Eruption	x^2 Waiting Time
1.983	59
5.1	96
2.367	63
1.8	53
1.667	64
4.05	81
4.117	79
2.133	67
4.7	88
3.5	87

https://kasumisanchika.medium.com/k-means-clustering-for-old-faithful-geyser-eruptions-analysis-74f3df8979b1

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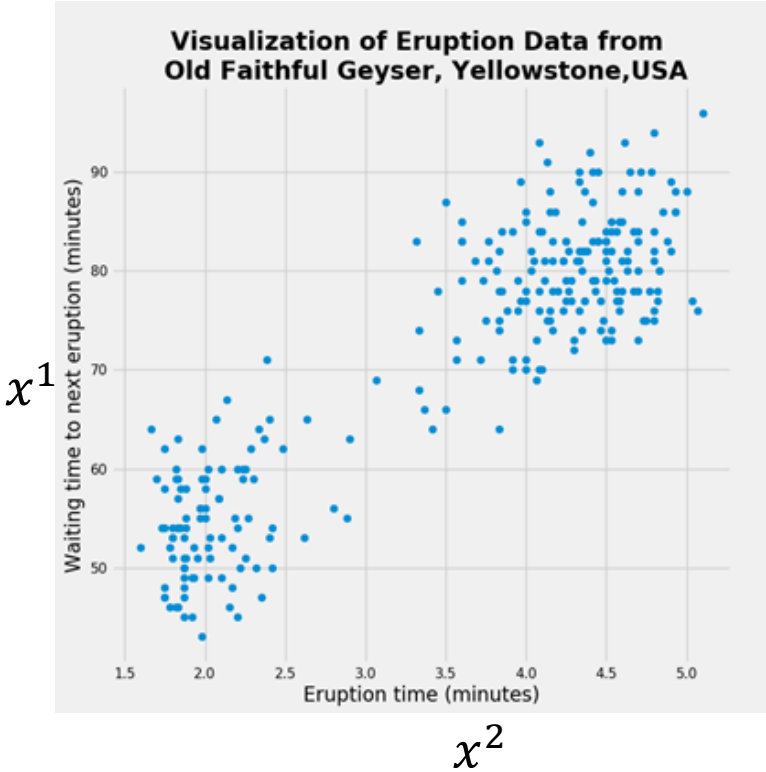

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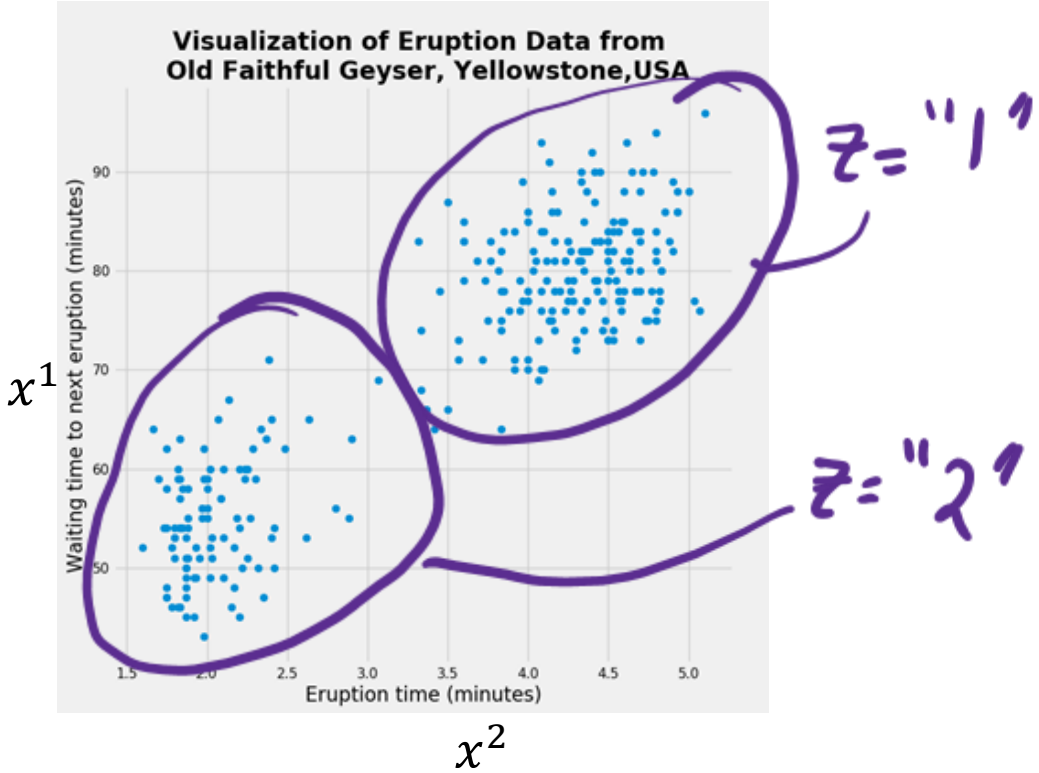

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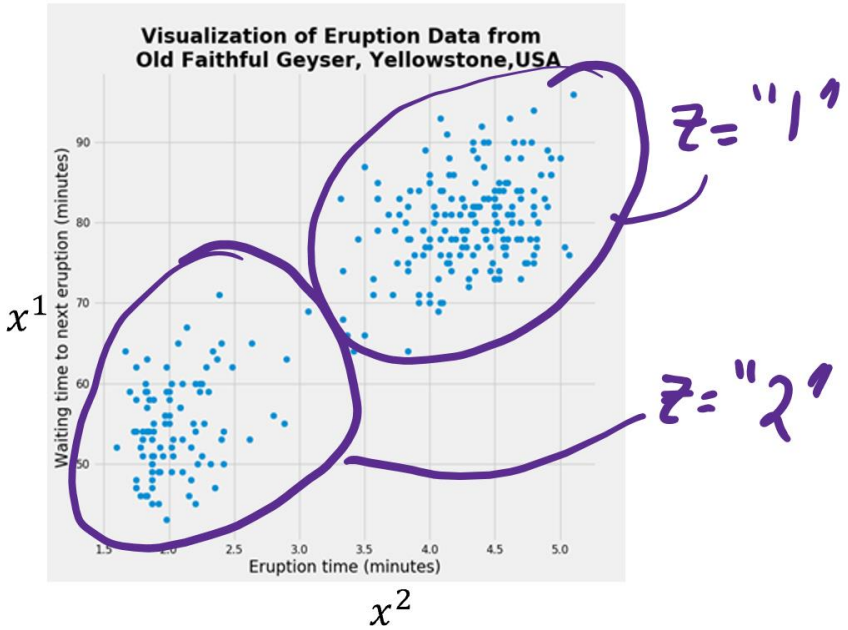
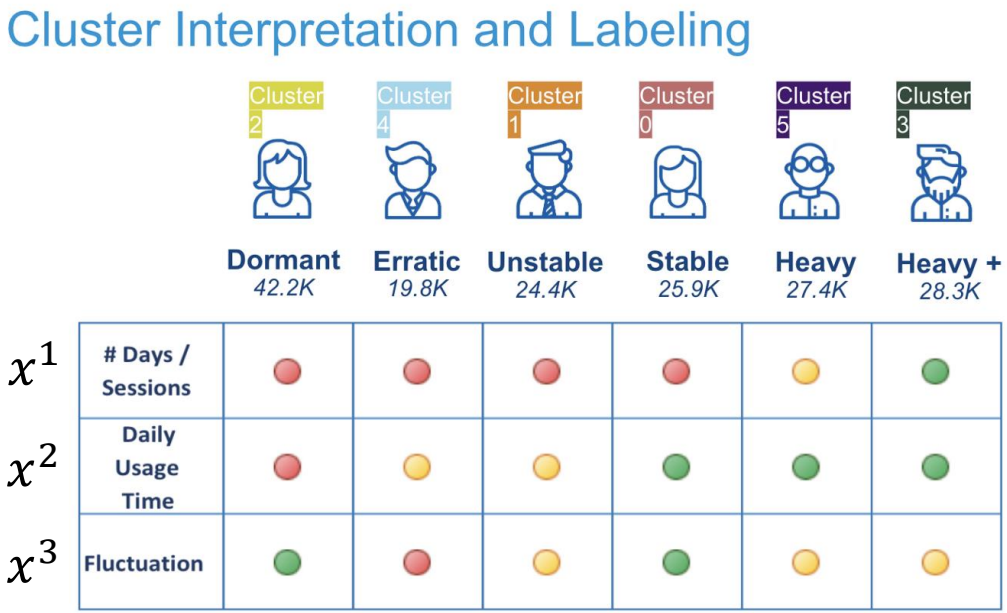
Clustering for data exploration

Clustering **finds hidden subgroups**:

- Patterns in nature (Geyser eruptions)
- Types of customers in a database from customer activities.
- Subtypes of disease for therapeutics from patient charts.
- Types of cells in a tissue from single cell data.
- Ancestry groups from genetic data.
- Finding topics in on-line documents.





https://medium.com/@sygong/k-means-clustering-for-customer-segmentations-a-practical-real-world-example-196a10323b9f

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i. Clustering, $z \in \mathbb{Z} = f_\theta(x)$.
ii. Dimensionality reduction, $z \in \mathbb{R}^m = f_\theta(x)$, $m \ll d$.
iii. Representation learning, $z \in \mathbb{R}^m$, $z = f_\theta(x)$, or $z \sim p_\theta(x)$.
iv. "Generative" modeling, $x \sim p_\theta(x)$.
v. Density estimation, estimate $p_\theta(x)$.

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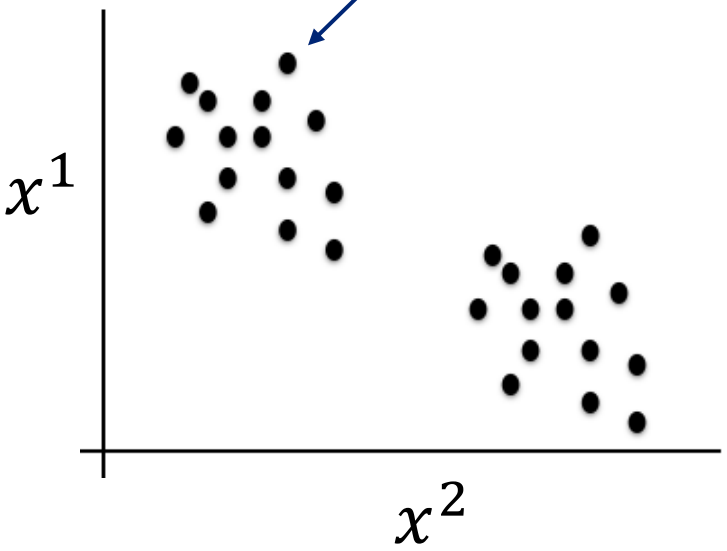
The main idea of *clustering* $\{x_i\}$

Suppose we had only input features, and no class labels:



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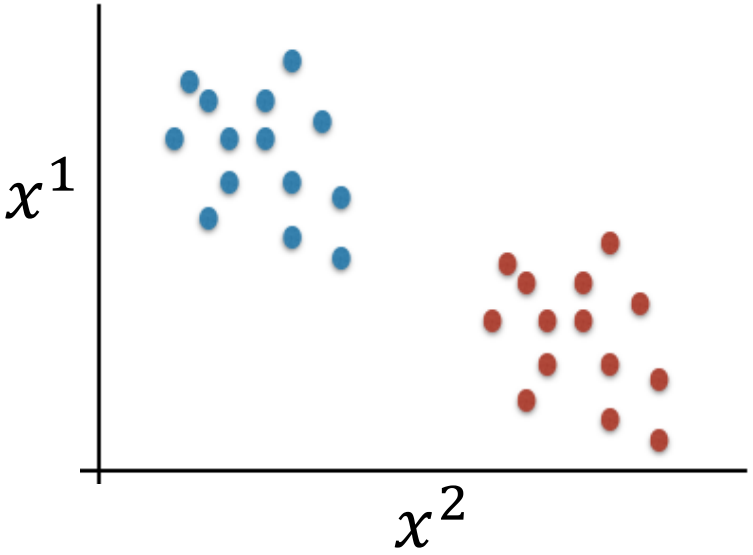

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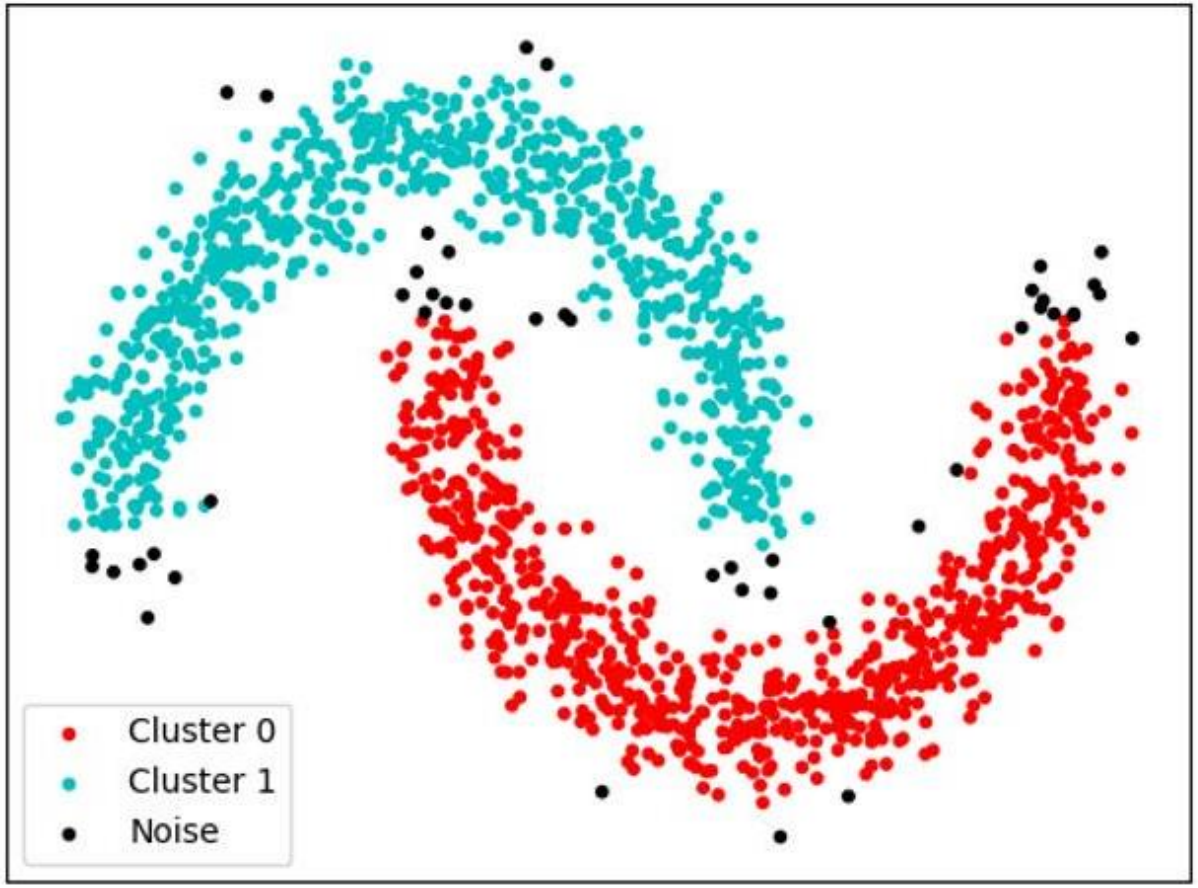
Clustering for outlier detection



https://www.imperva.com/blog/2017/07/clustering-and-dimensionality-reduction-understanding-the-magic-behind-machine-learning/

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Three broad approaches to clustering

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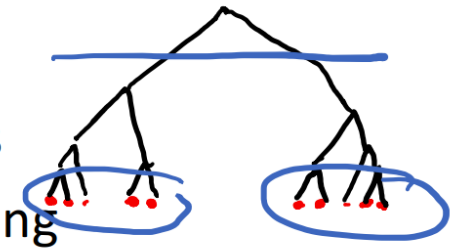
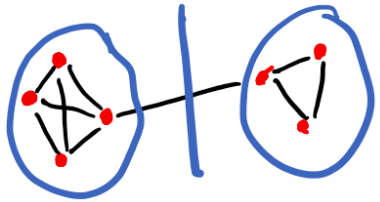
- Hierarchical clustering
  - Build a tree (bottom-up or top-down), representing distances among data points
  - **Example**: single-, average- linkage clustering



- Partitional approaches
  - Define and optimize a notion of "cost" defined over partitions
  - **Example**: Spectral clustering, graph-cut based approaches



- Model-based approaches
  - Maintain cluster "models" and infer cluster membership (e.g., assign each point to closest center)
  - **Example**: k-means, Gaussian mixture models, …

[slide from Andrea Krause]

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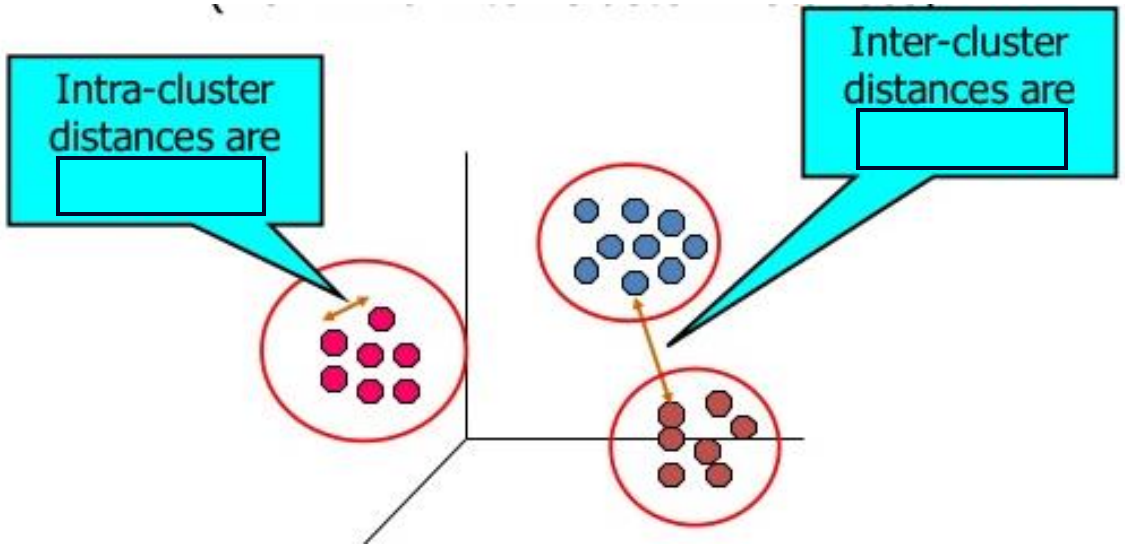
Main desiderata of clustering

1. Want high intra-cluster similarity.
2. Want low inter-cluster similarity.



3. Similarity/distance is in the eye of the beholder!

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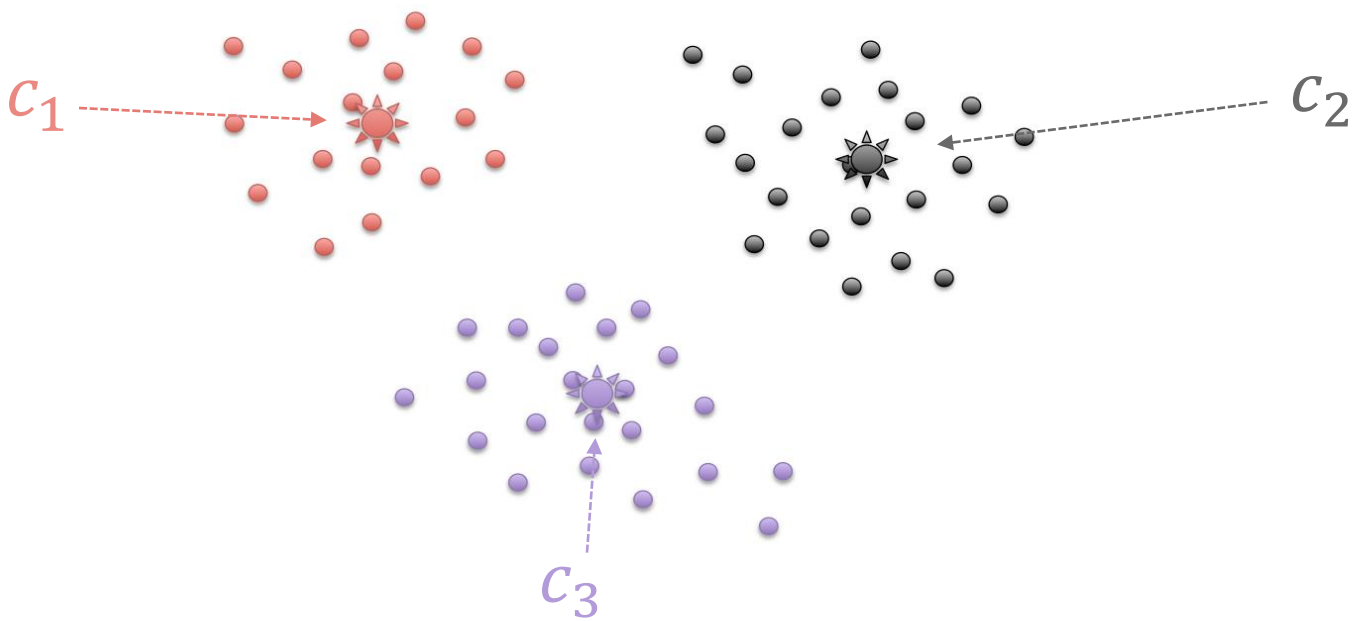
https://www.slideshare.net/NontawatB/08-clustering

Centroid-based clustering

- Each cluster is represented by a point in the input space: a *centroid*, $c_k \in R^d$ (for $X \in R^d$).

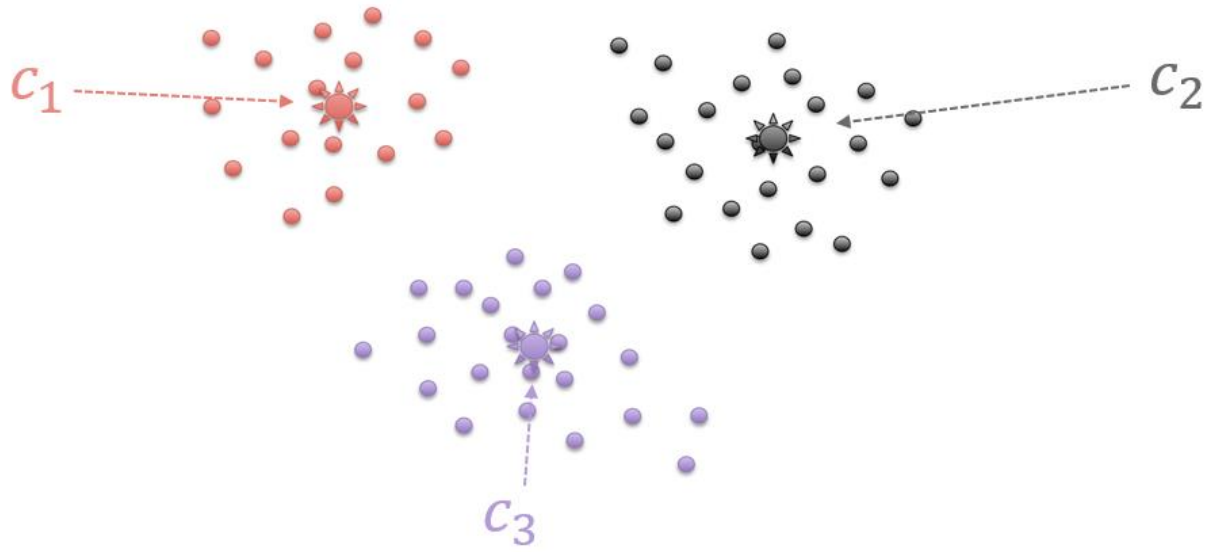


- "K-means" is the most common centroid-based approach.

Portions of some slides courtesy of Yisong Yue at Caltech.

K-means clustering

- Model parameters are $\{c_k\}$.
- Chosen such that:
  - the distance of each point, x_i, to its assigned centroid, is minimized.



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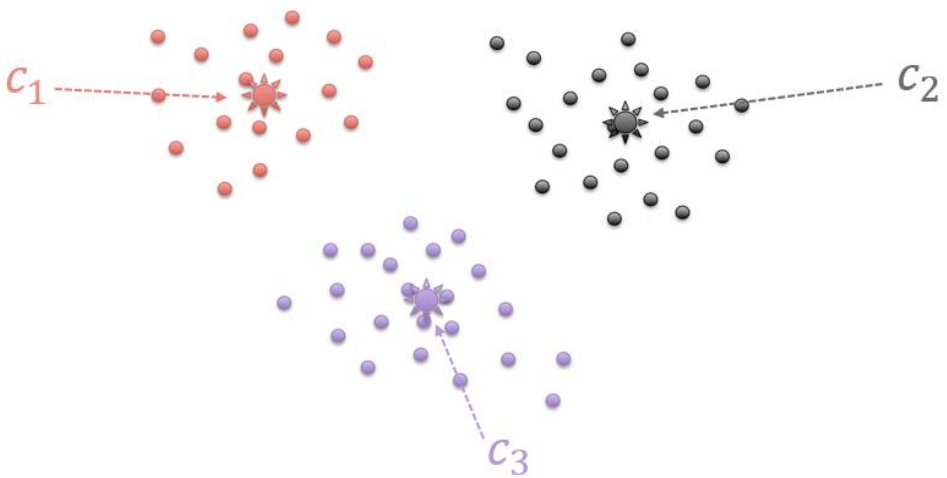
Formally: K-means clustering

- Input: training data, $X = \{x_i\}_{i=1}^{n}, x_i \in R^d$.
- Parameters are $\{c_k \in R^d\}$.
- A cluster partition of the data, $C_1 \cup C_2 \cup \cdots C_K$, wherein every x_i is assigned to one (and only one) of the K clusters.
- Objective/loss function, and resulting optimization problem:

$$\underset{\underbrace{S=C_1 \square \ldots \square C_K}_{\text{cluster partition}},\ \underbrace{\{c_1,\ldots,c_K\}}_{\text{cluster centroids}}}{\text{argmin}} \square_k \square_{x \square C_k} \left\| x - c_k \right\|^2$$



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Parameter learning in K-means

- Suppose we knew $C_1 \cup C_2 \cup \cdots C_K$ how could we find $\{c_k\}$?

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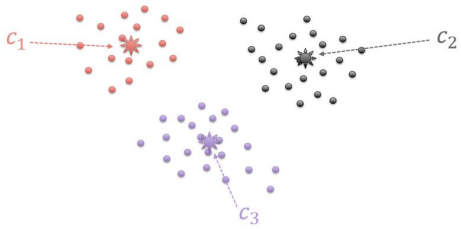

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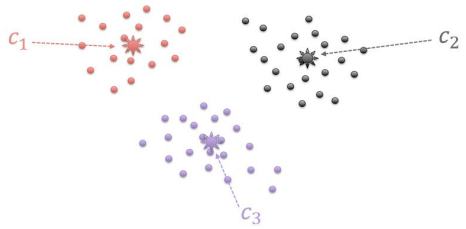

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$$\sum \frac{\partial}{\partial c_k} (x - c_k)^T (x - c_k)$$
$$= -2 \sum (x - c_k)$$
$$= 0 \Rightarrow \sum x = \sum c_k$$

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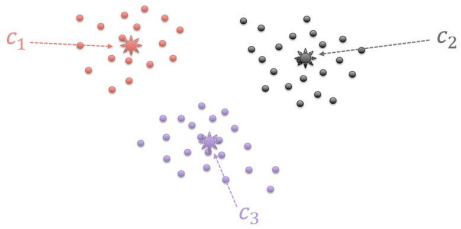

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- For which the answer is $\hat{c}_k = \frac{1}{N} \sum_{x \in C_k} x$

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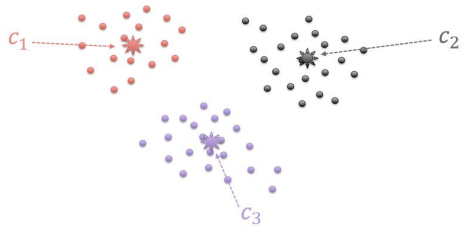

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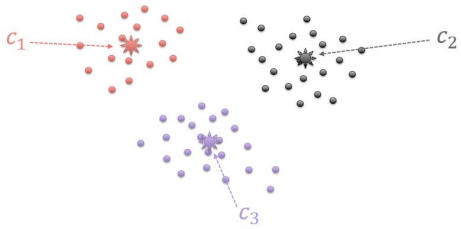
Other way around (parameter learning)

- Suppose we knew $\{c_k\}$, how could we find $C_1 \cup C_2 \cup \cdots C_K$?
- Answer: choose the cluster which is closest to each point, $z_i \equiv \underset{k}{\operatorname{argmin}} \|x_i - c_k\|^2$, and then $\hat{C}_k = \{x_i | z_i = k\}$.

$$\underset{S=C_1 \square \ldots \square C_K,\ \{c_1,\ldots,c_K\}}{\operatorname{argmin}} \square_k \square_{x \square C_k} \|x - c_k\|^2$$



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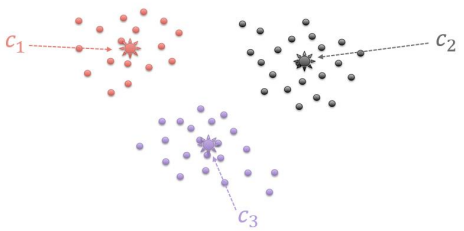
The K-Means algorithm ("Lloyd's Algorithm")



$$\underset{S=C_1 \cup \ldots \cup C_K,\ \{c_1,\ldots,c_K\}}{\operatorname{argmin}} \sum_k \sum_{x \in C_k} \|x - c_k\|^2$$

1. Initialize the cluster centers, $\{c_k\}$

   (e.g., pick k points at random from your training data).

2. Repeat until convergence:
   i. Compute partition $C_1 \cup C_2 \cup \cdots C_K$, given the $\{c_k\}$.
   ii. Compute centers $\{c_k\}$, given $C_1 \cup C_2 \cup \cdots C_K$.

Does this converge?

- Yes: at each step, we are reducing the objective function or have converged.
- If assignments do not change, we have a <u>local</u> min.

Can you think of a slight variation to this algorithm that's also valid?

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The K-Means algorithm

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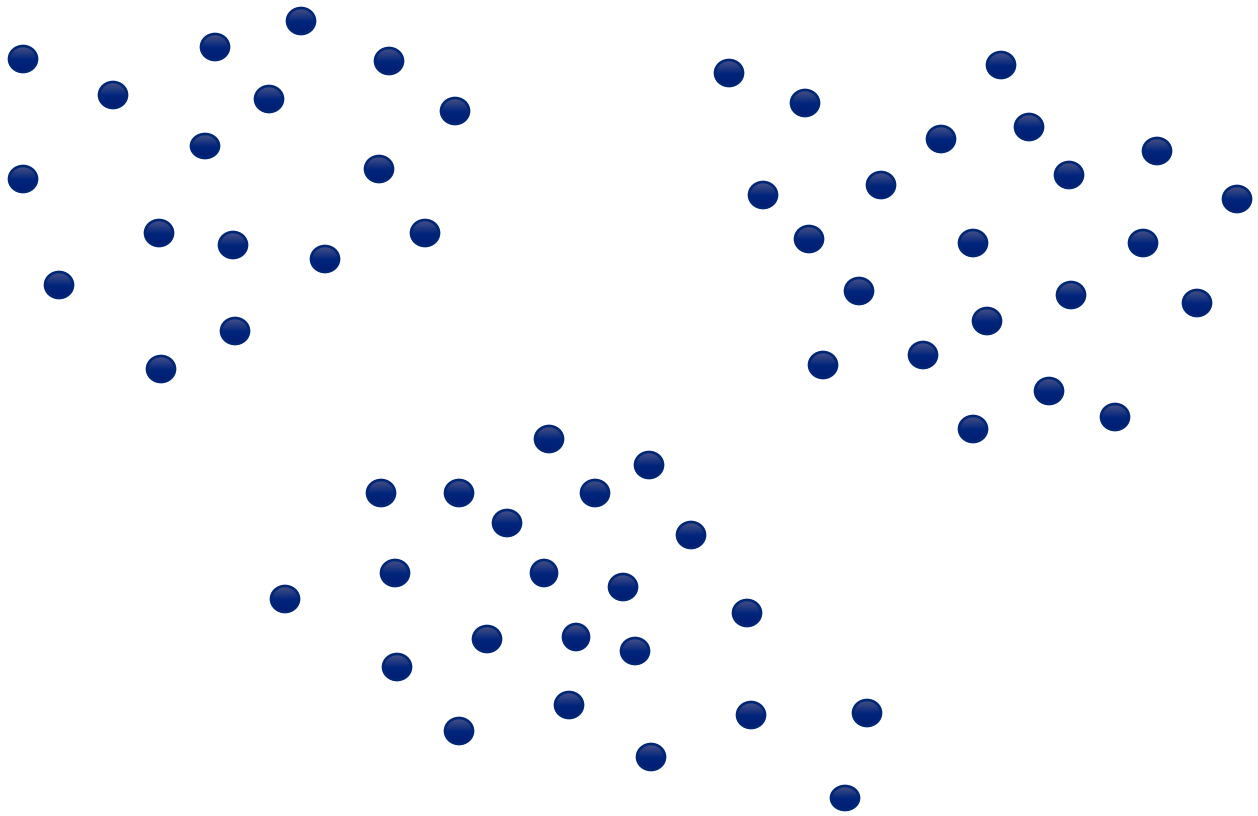

[slide courtesy Yisong Yue]

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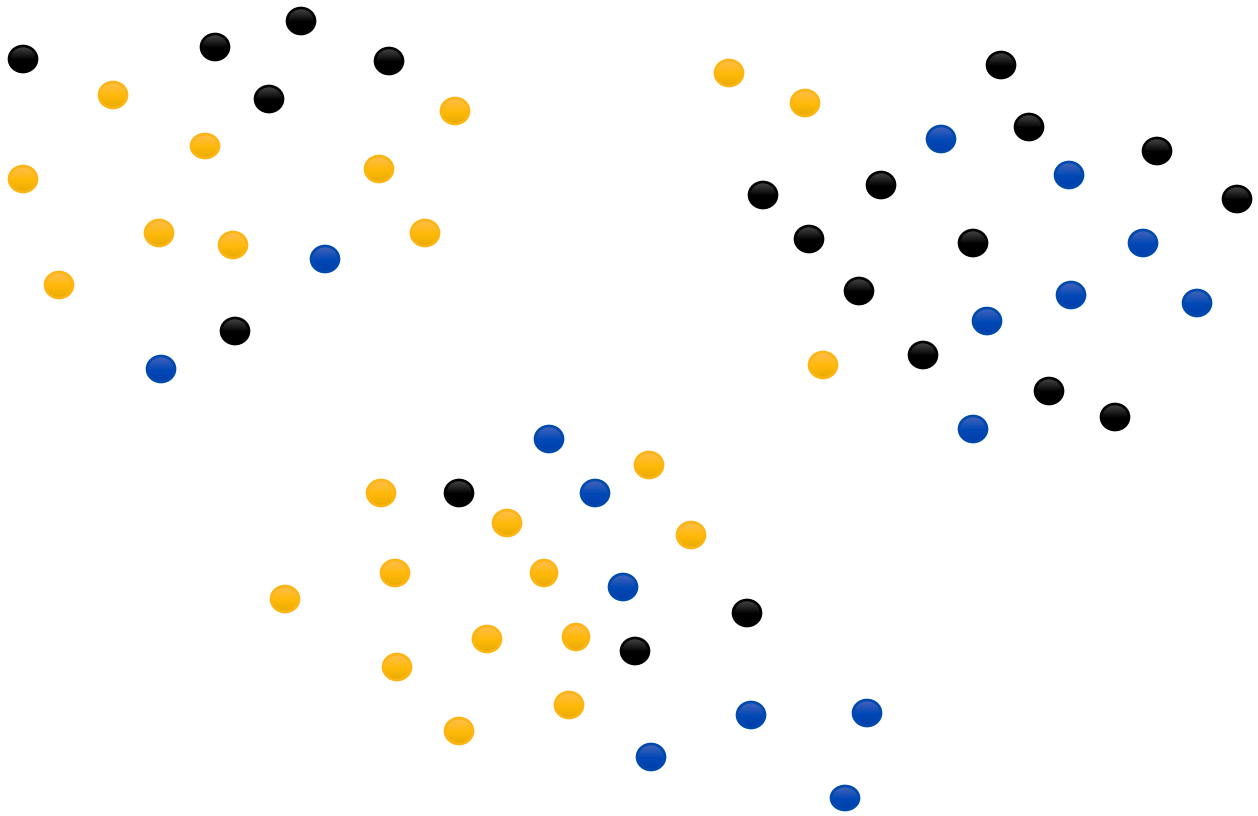

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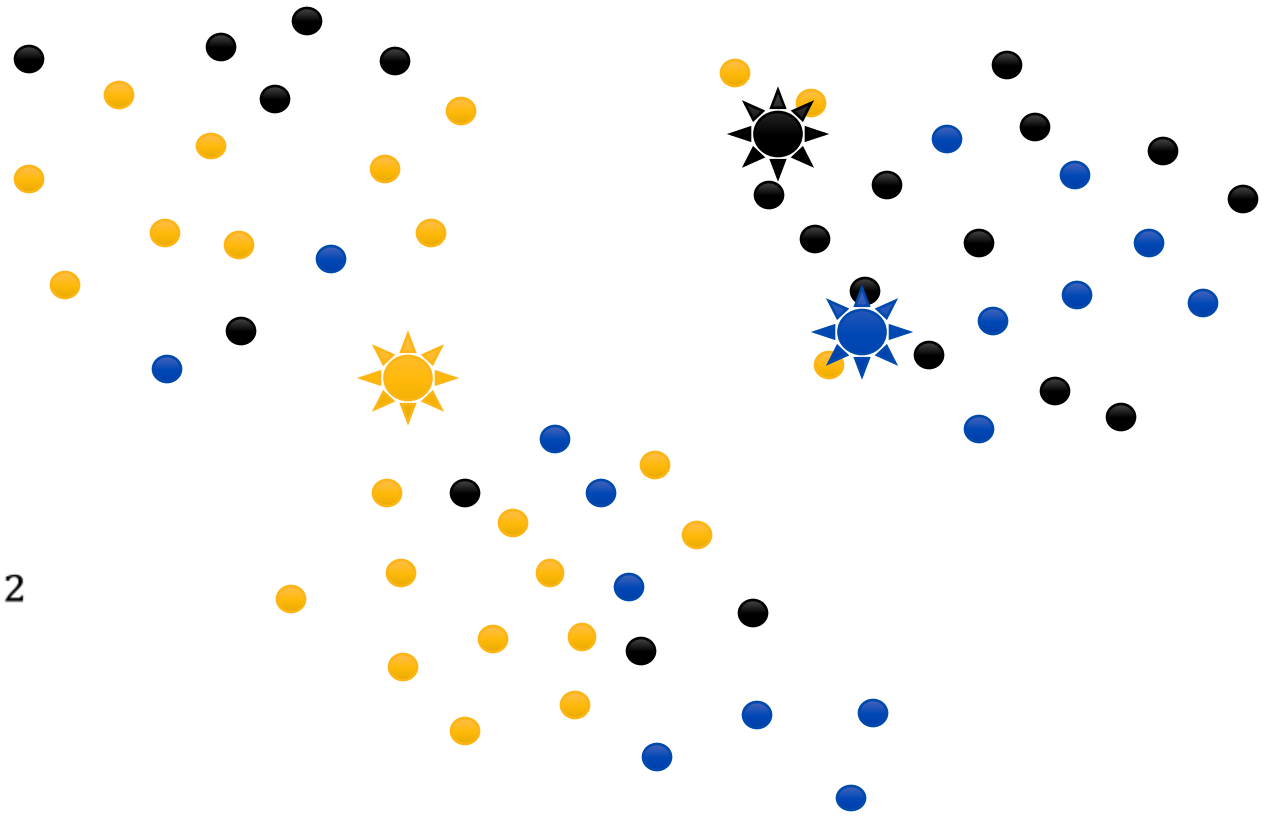

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[slide courtesy Yisong Yue]

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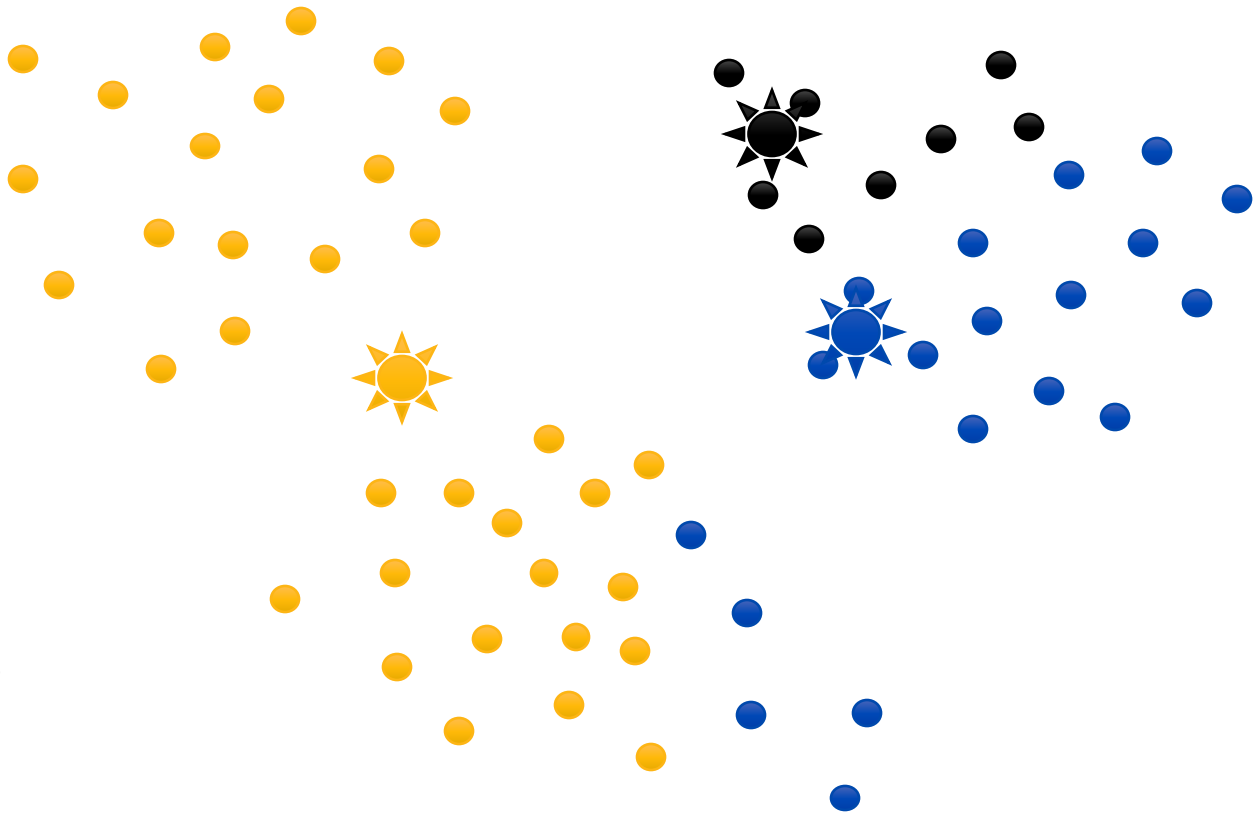

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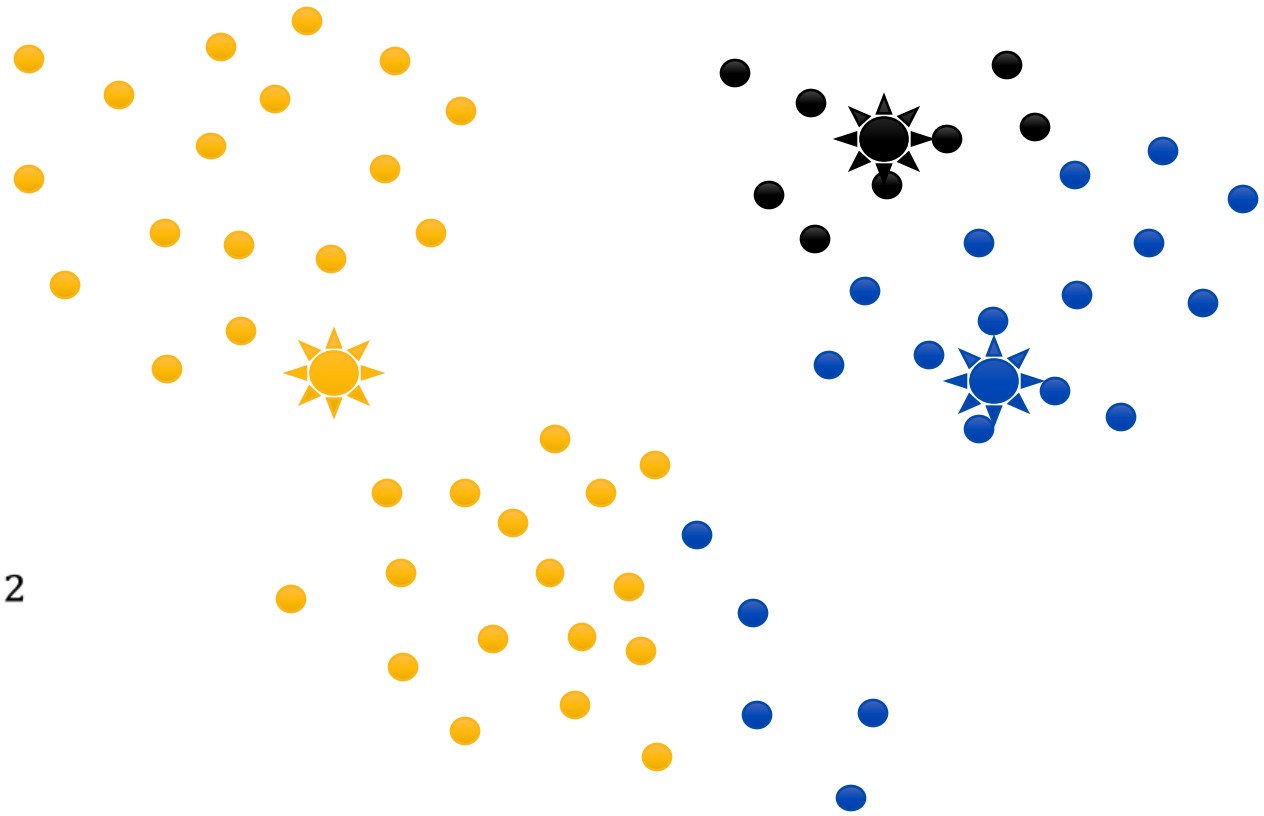

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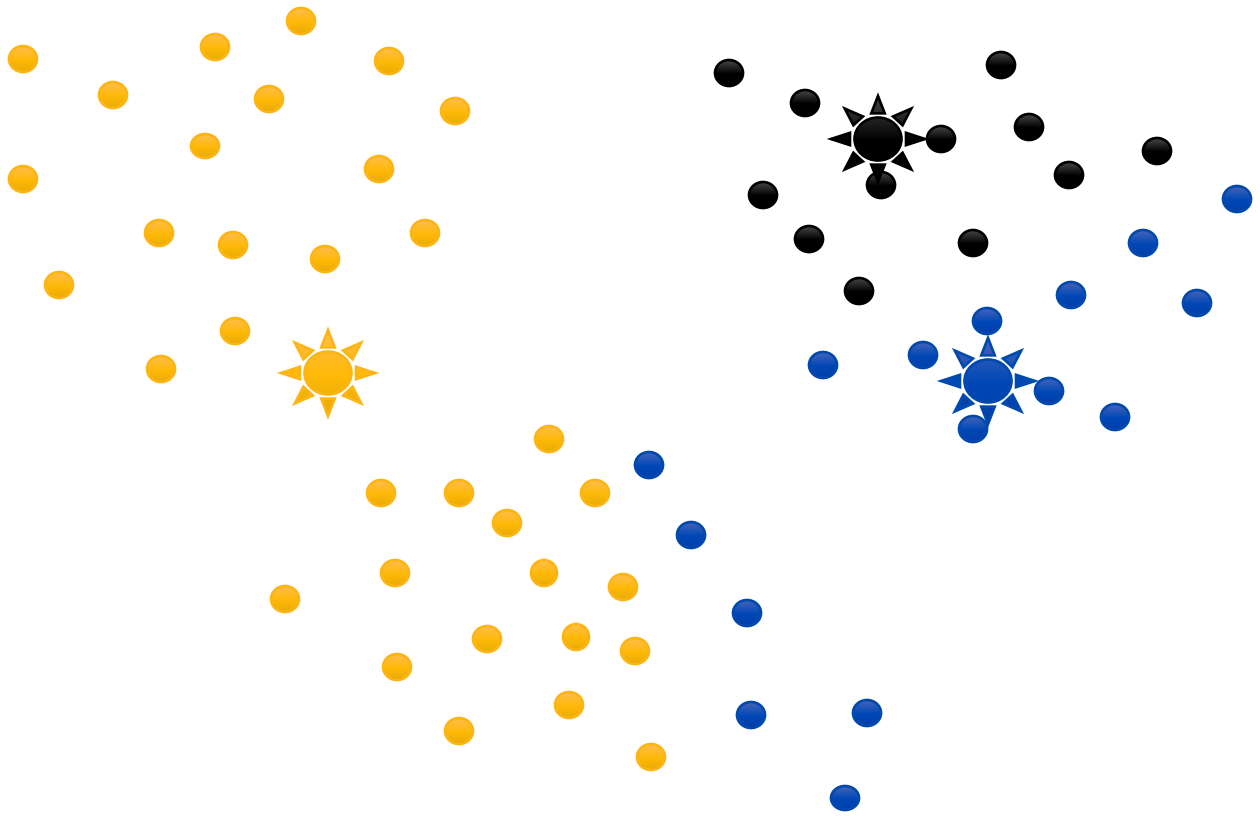

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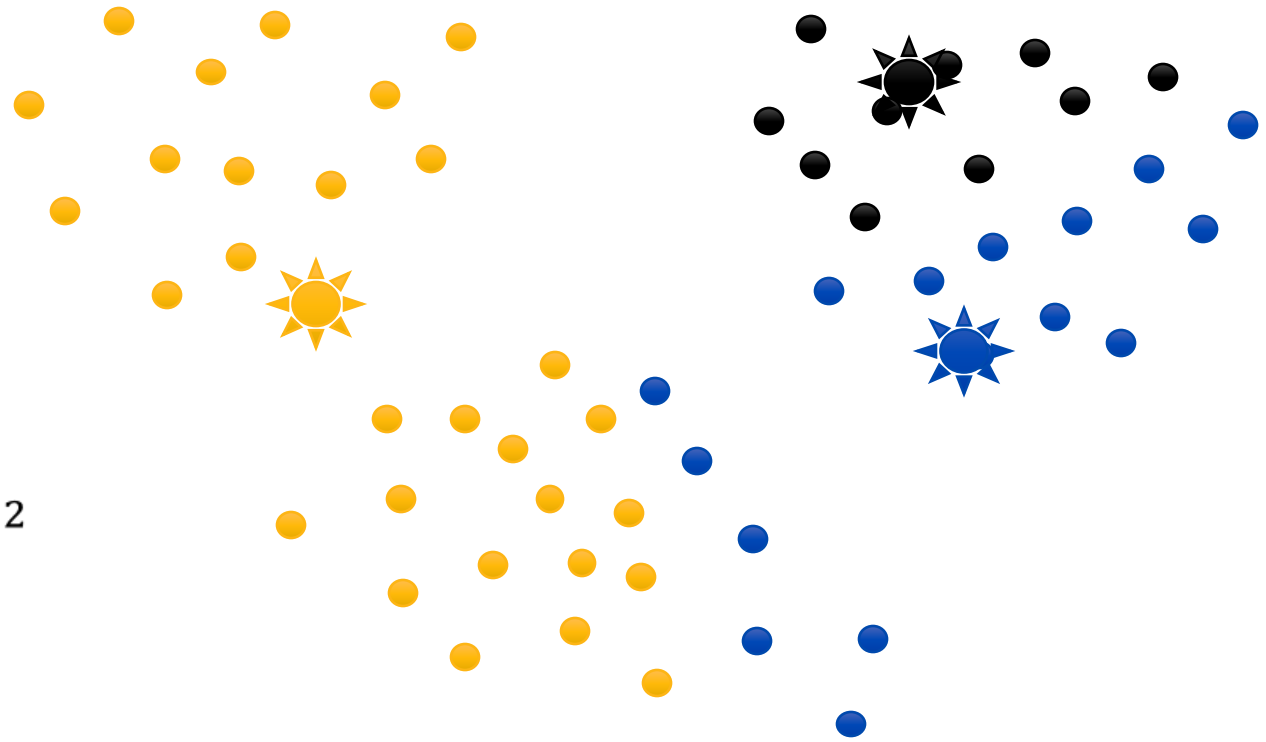

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The K-Means algorithm

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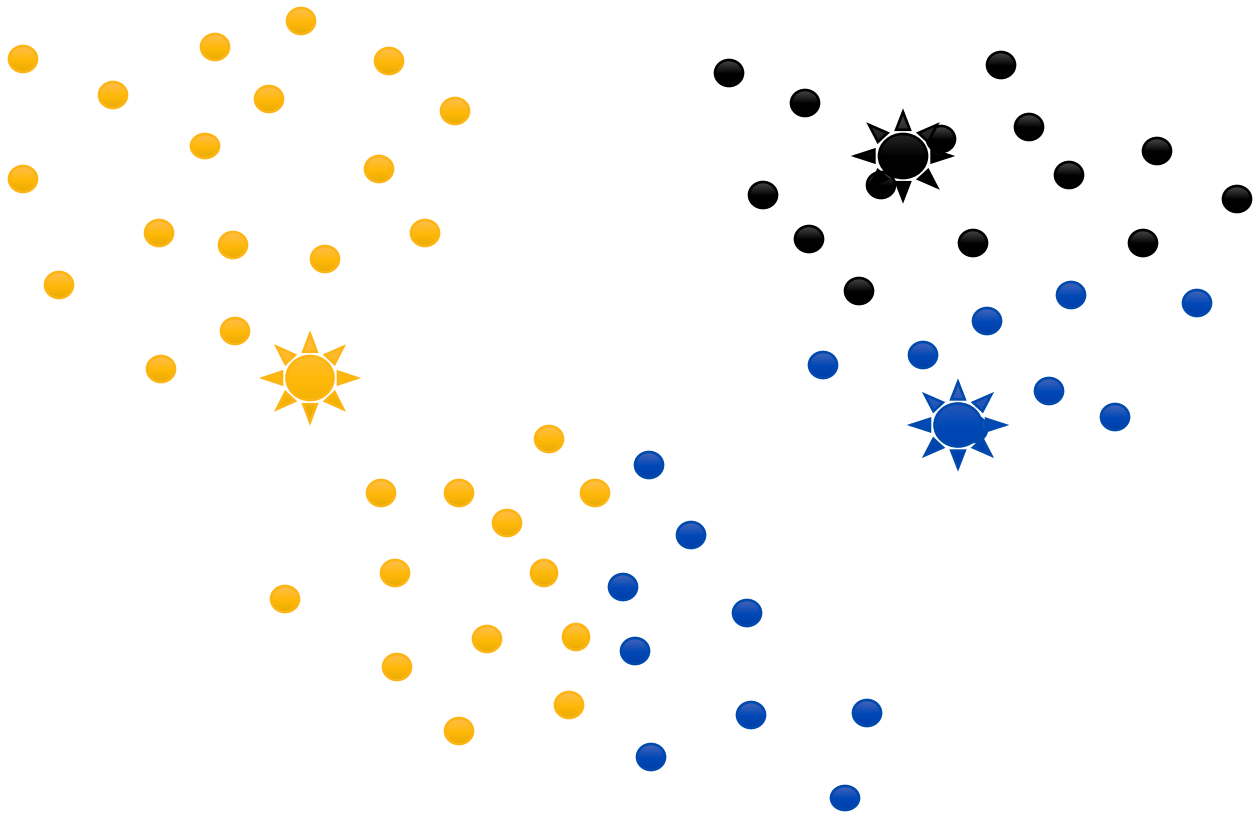

$$z_i \equiv \underset{k}{\operatorname{argmin}} \|x_i - c_k\|^2$$

$$\Rightarrow \hat{C}_k = \{x_i | z_i = k\}.$$

[slide courtesy Yisong Yue]

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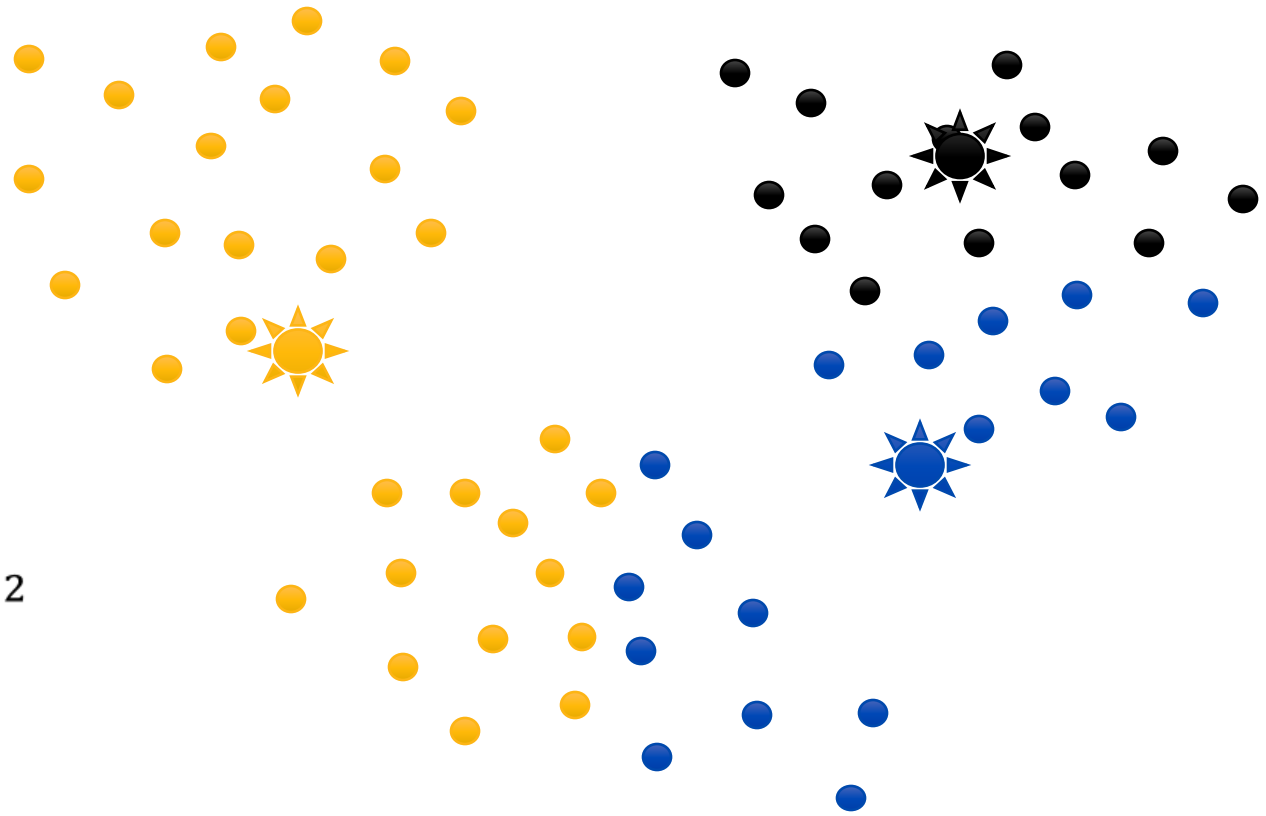

$$\hat{c}_k = \underset{c_k}{\operatorname{argmin}} \sum_{x \in C_k} \|x - c_k\|^2$$

$$\Rightarrow \hat{c}_k = \frac{1}{N} \sum_{x \in C_k} x$$

[slide courtesy Yisong Yue]

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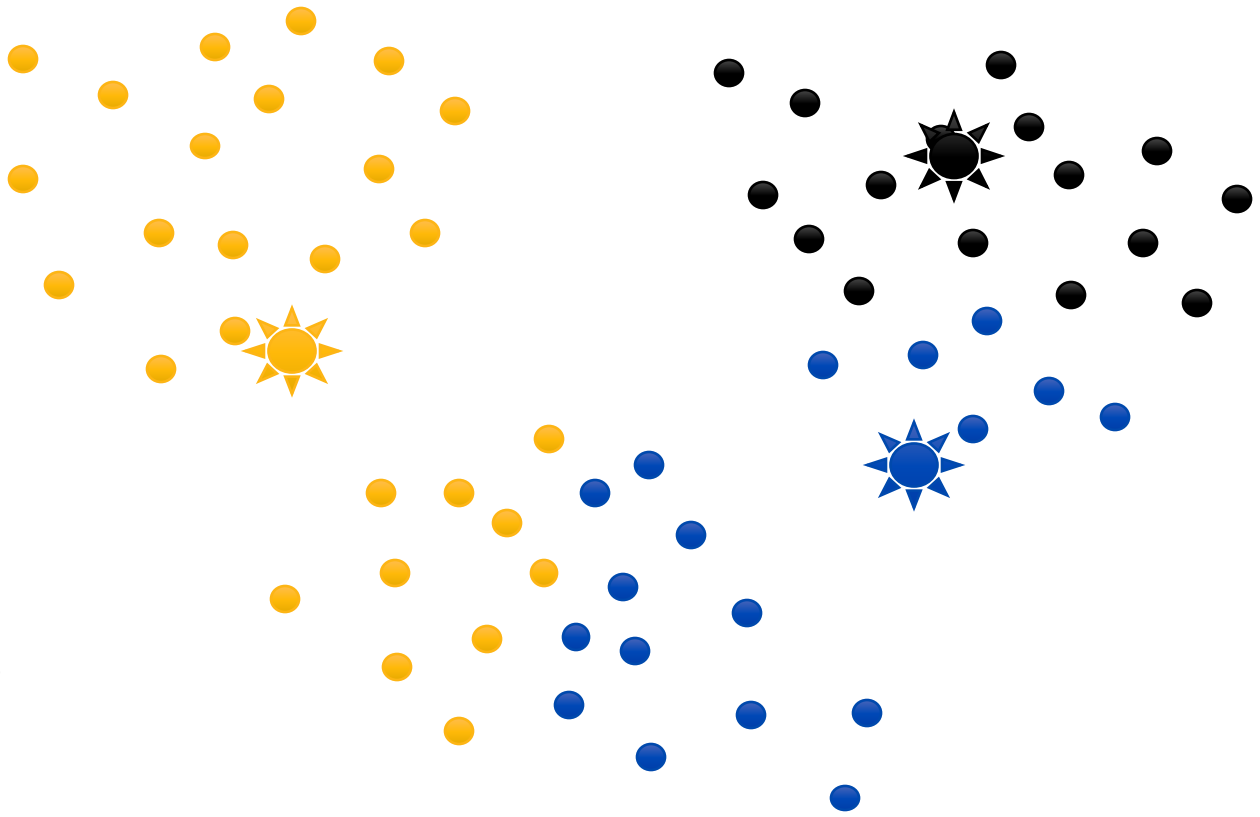

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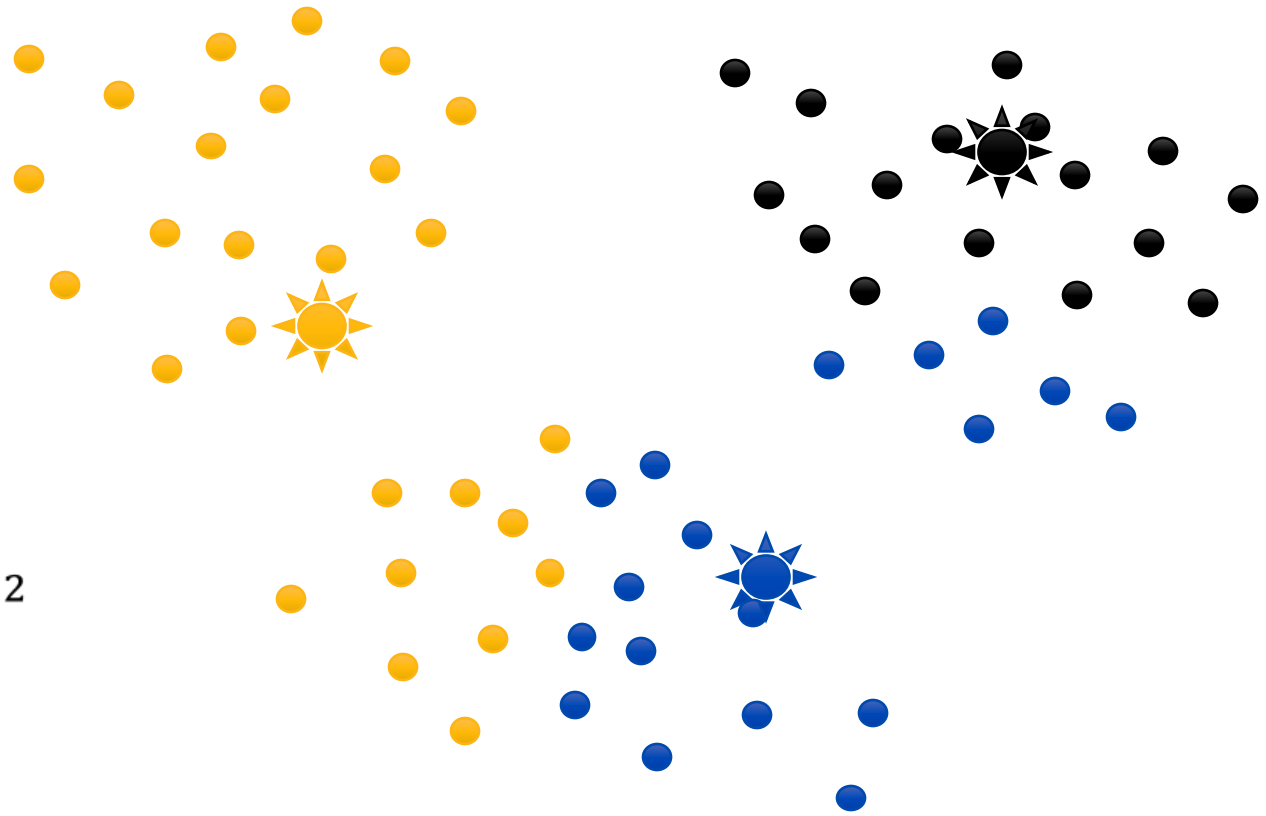

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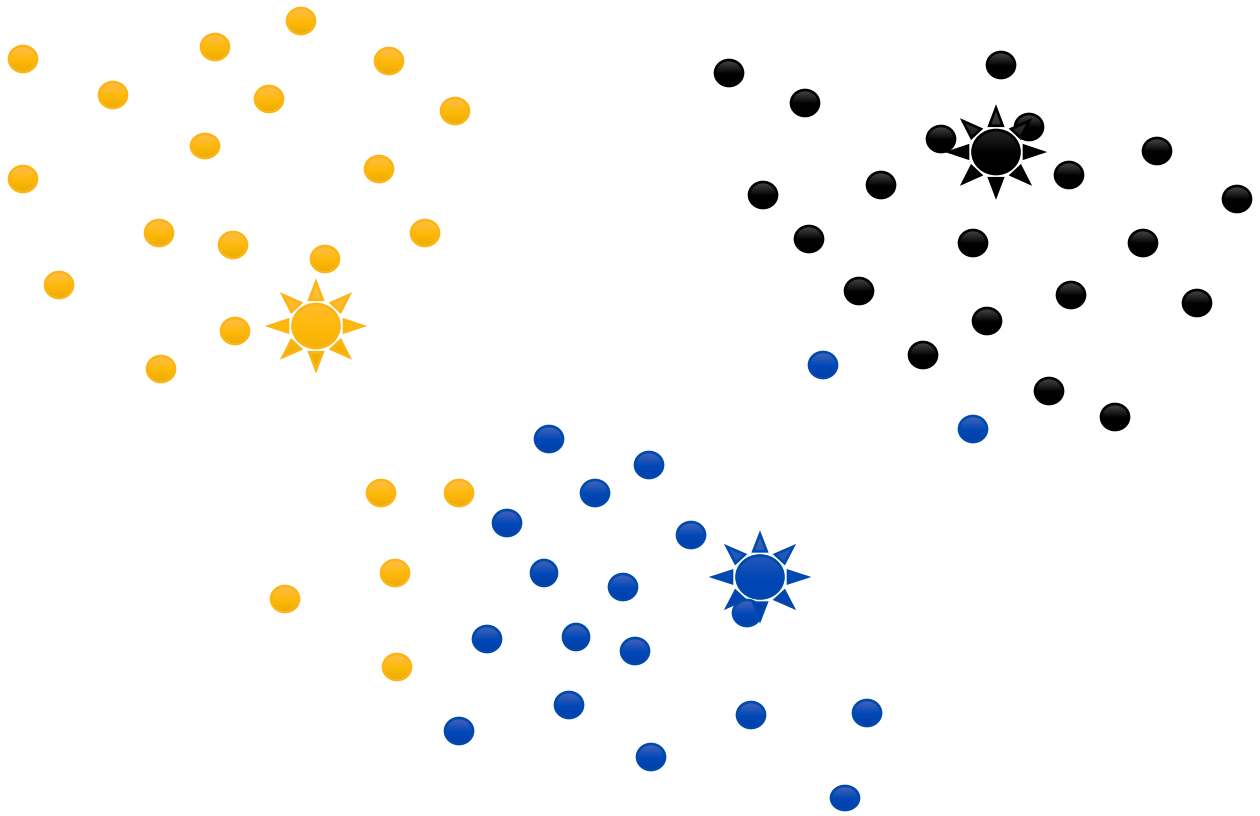

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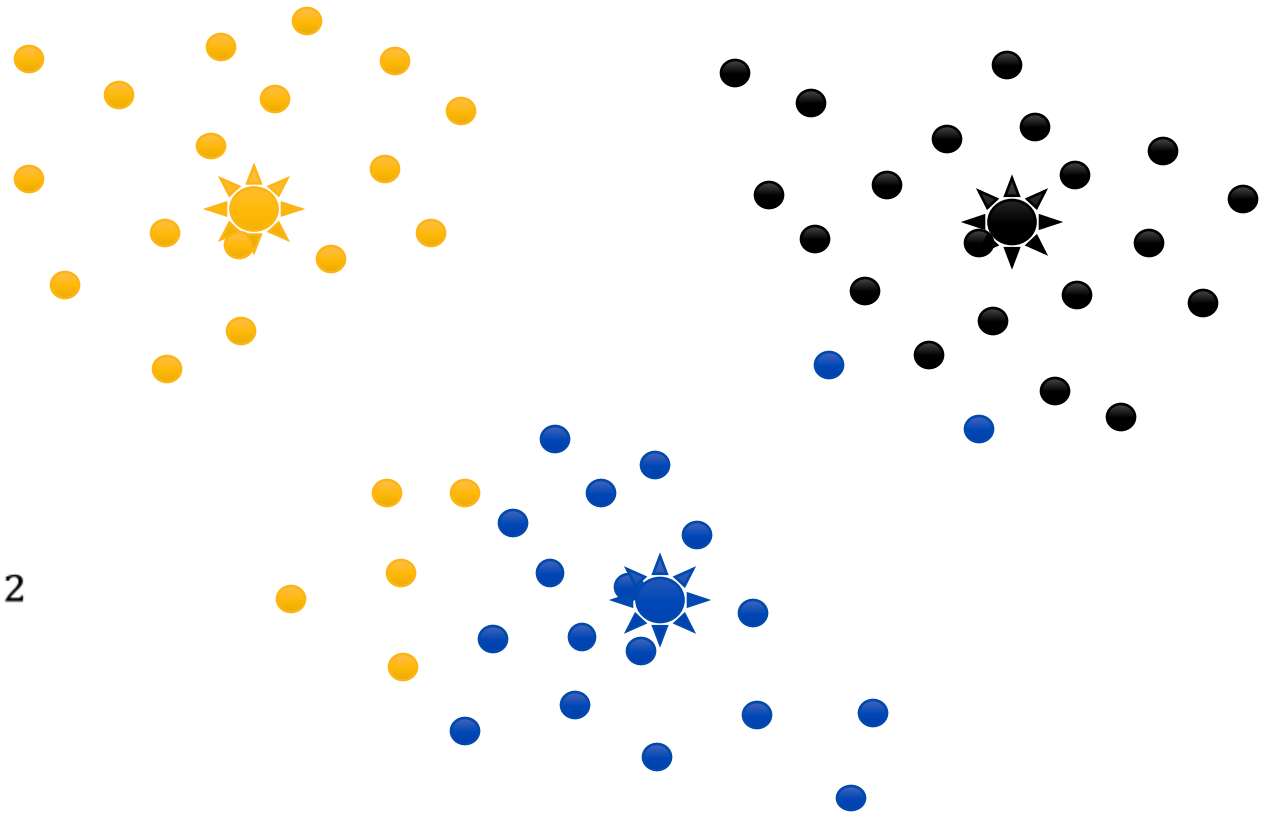

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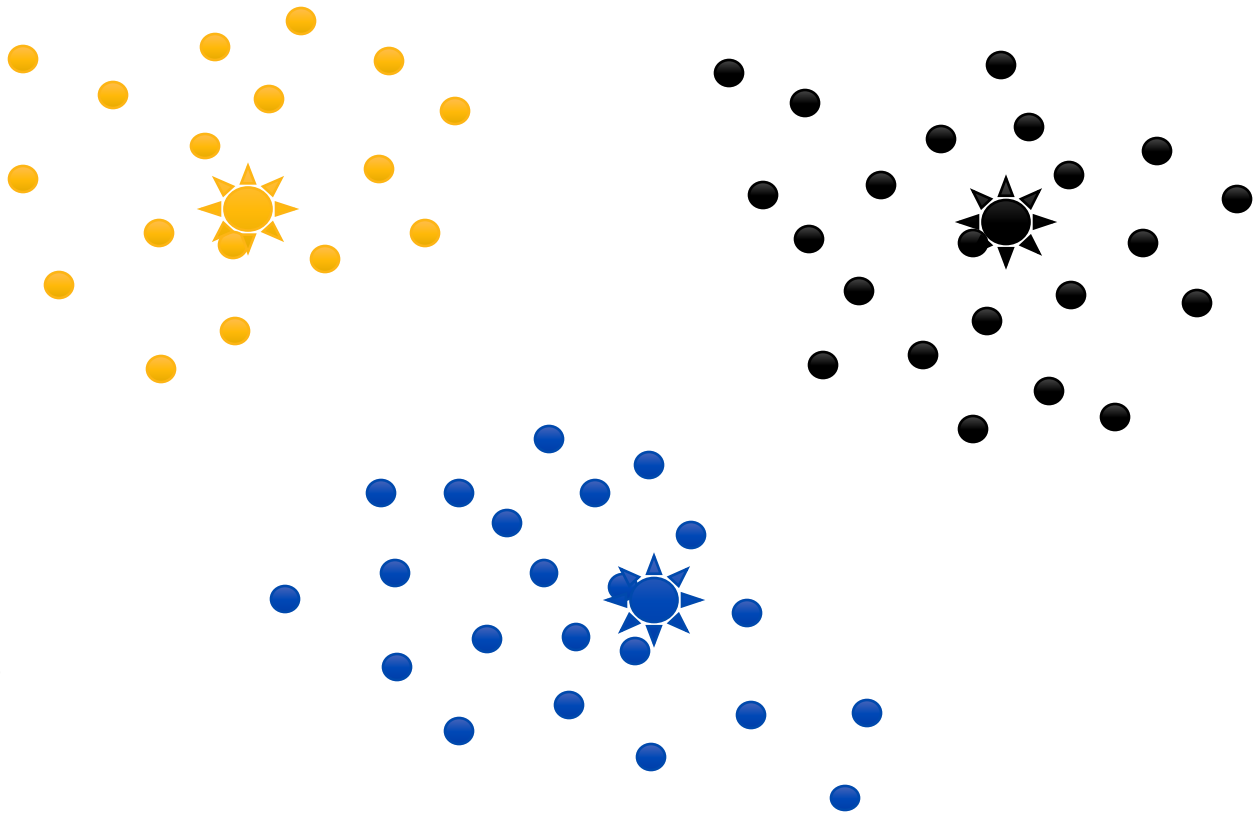

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[slide courtesy Yisong Yue]

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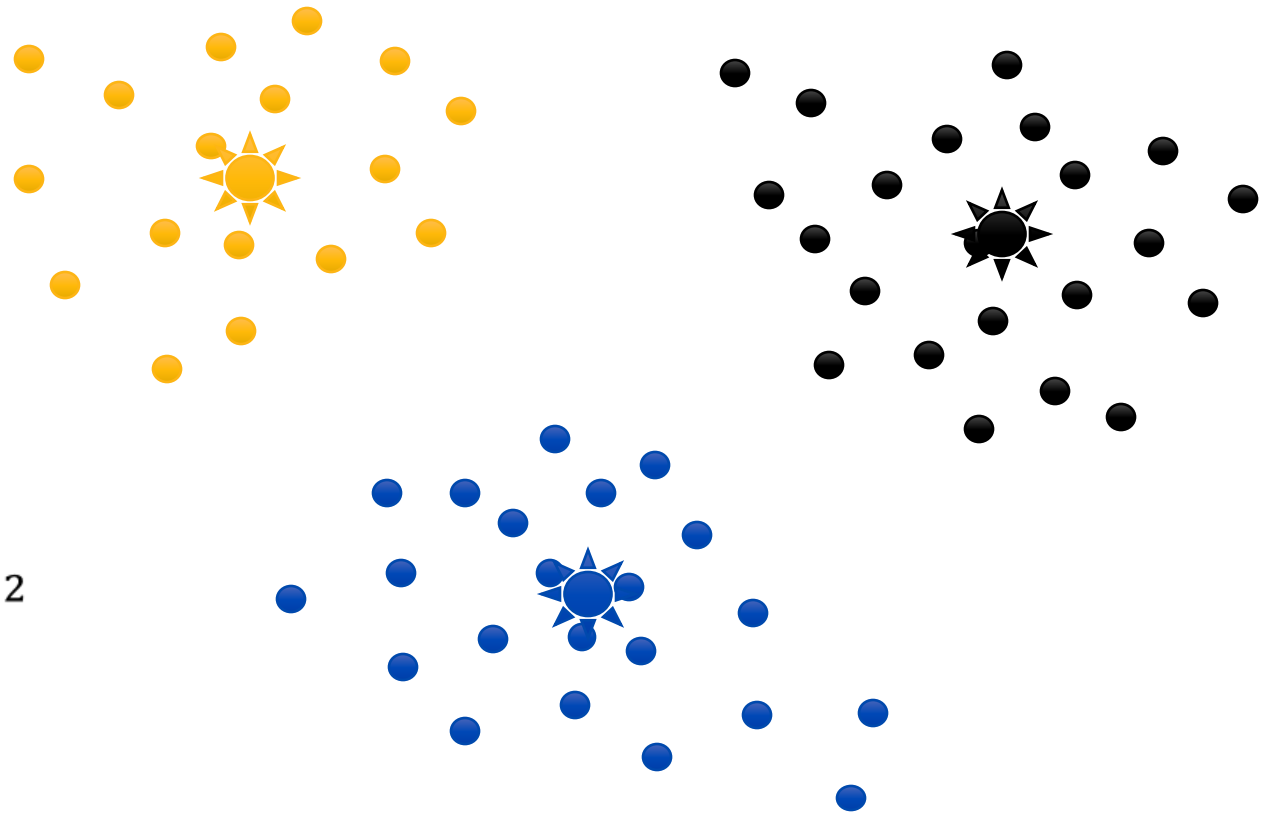

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[slide courtesy Yisong Yue]

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Aside: distances, metrics and similarities.

- "want points to be similar/dissimilar" within/across clusters
- "want distance to be minimized/maximized".

Properties of a distance function (metric):

1. $j = k$ iff $d(j,k) = 0$.
2. $j \neq k$ iff $d(j,k) > 0$.
3. *symmetry,* $d(j,k)=d(k,j)$ *(why KL-divergence is not a distance)*
4. *triangle inequality,* $d(i,j) + d(i,k) \geq d(j,k)$

Is K-means "distance" a metric? $\hat{c}_k = \underset{c_k}{\operatorname{argmin}} \sum_{x \in C_k} \|x - c_k\|^2$

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Properties of a **distance** function (metric):

1. $j = k$ iff $d(j,k) = 0$.
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3. *symmetry,* $d(j,k)=d(k,j)$ *(why KL is not a distance)*
4. *triangle inequality,* $d(i,j) + d(i,k) \geq d(j,k)$

dissimilarity: intuitively related, but may not satisfy metric properties.
similarity: "complement" of dissimilarity, e.g.:

$$similarity(j,k) = 1 - dissimilarity(j,k)$$

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How to find good # of clusters?

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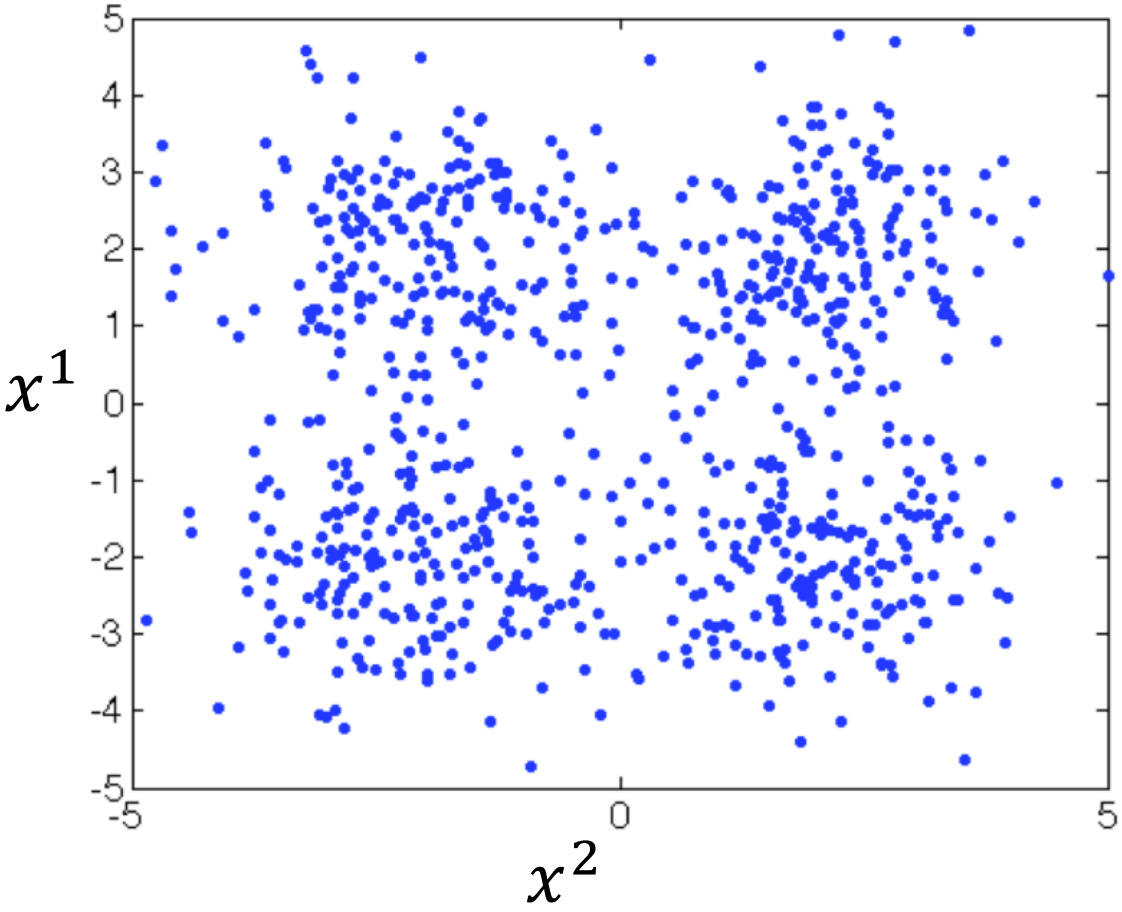

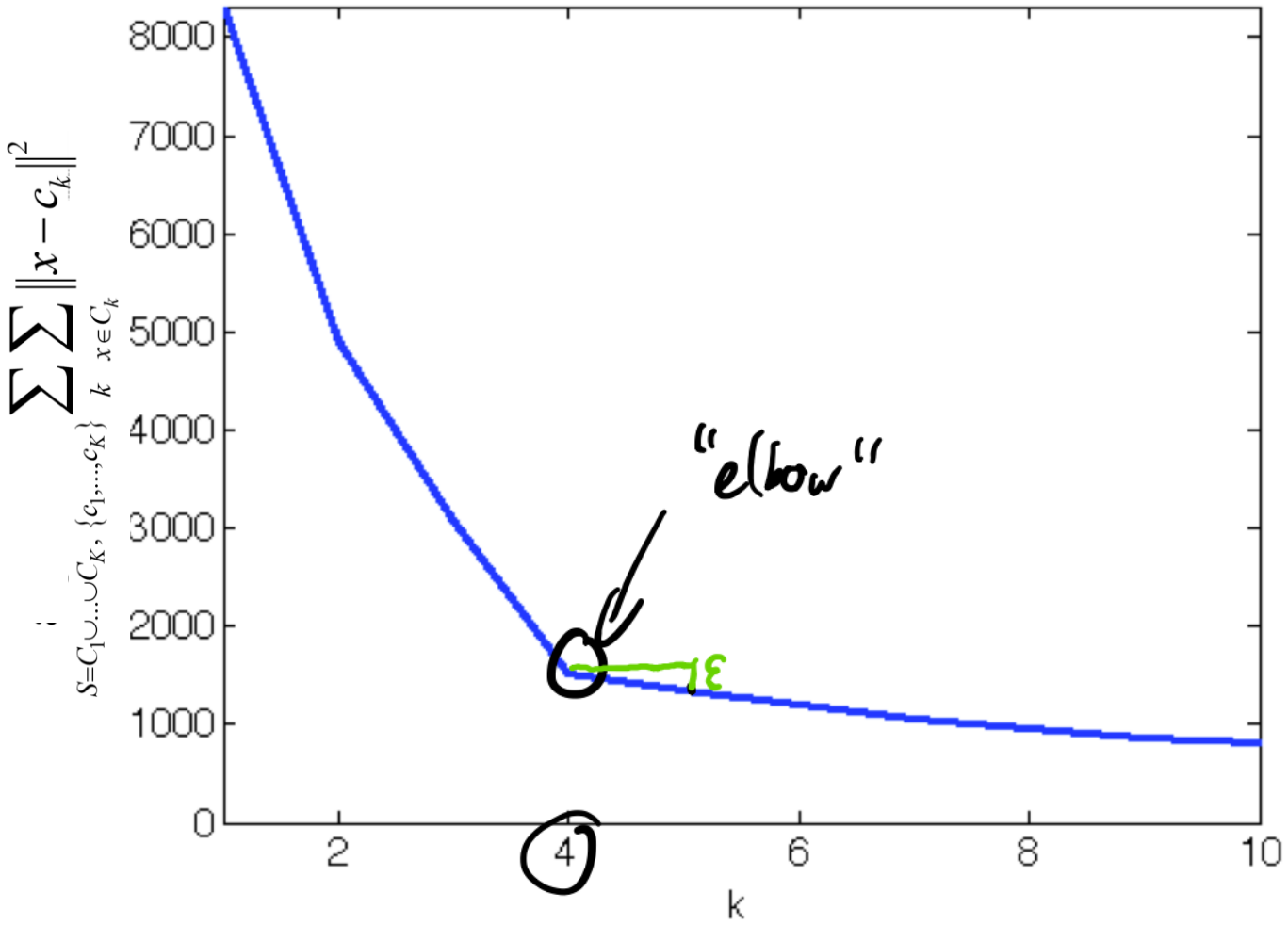

[slide from Andrea Krause]

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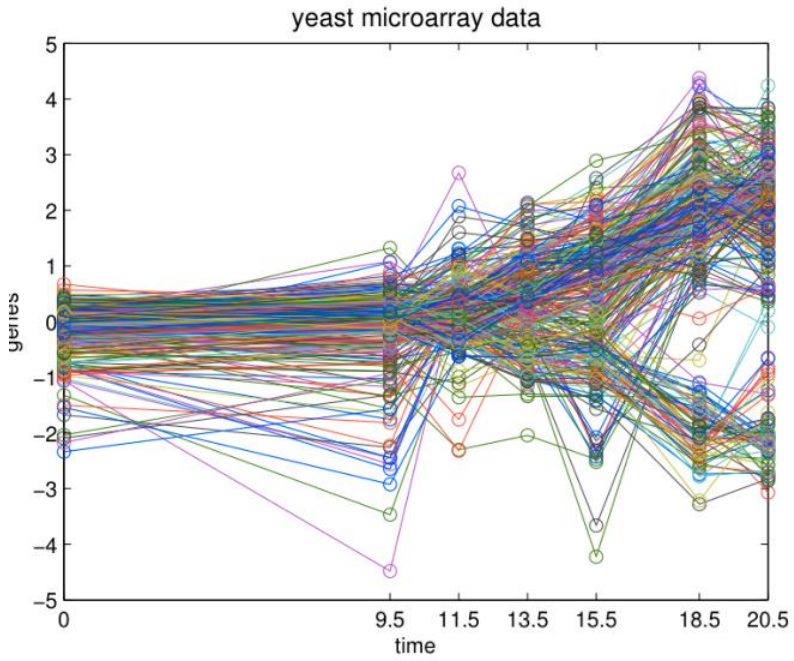
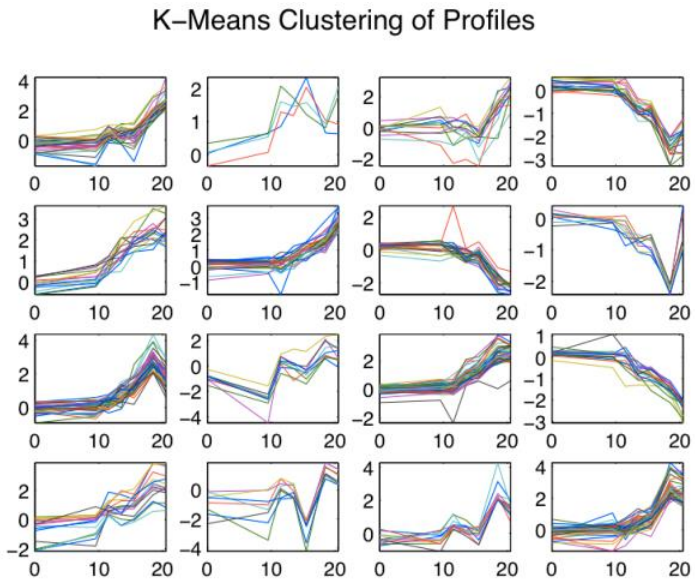
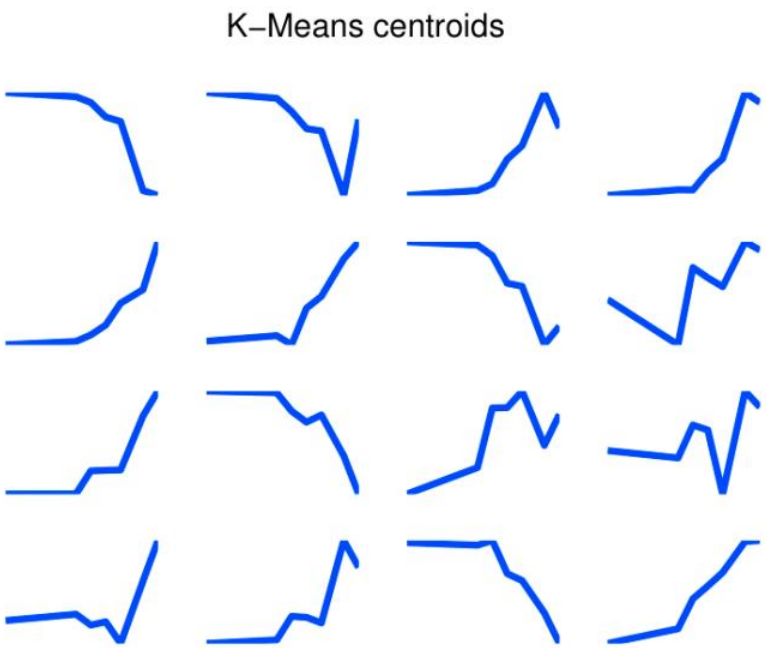
Application of K-Means Clustering







clustering yeast genes by their "gene expression" measurements over time

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http://www.cs.toronto.edu/~urtasun/courses/CSC2515/CSC2515_Winter15.html

Pixel K-Means

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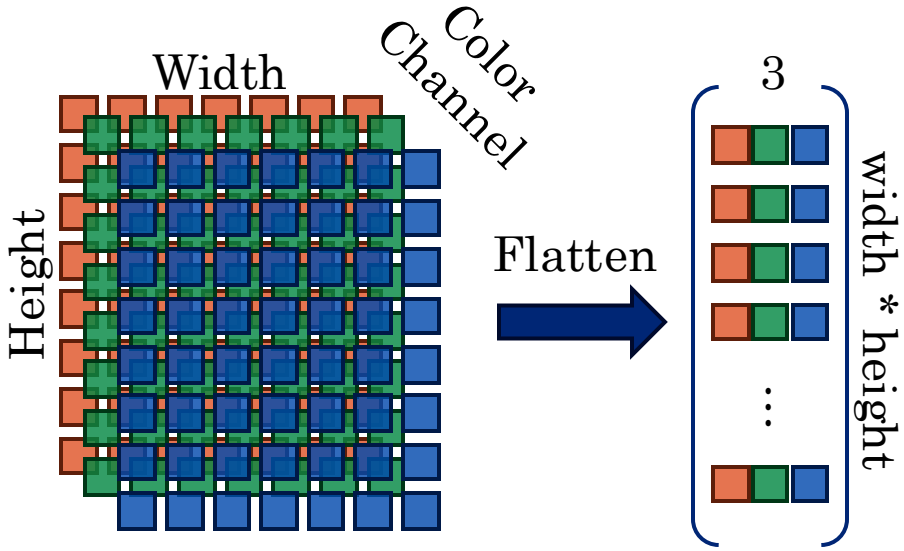
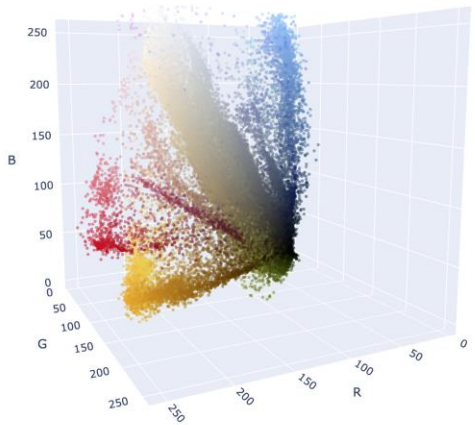
The pixels in an image can be treated of as **vectors** in an **RGB vector space**.

- We can use k-means to compute the **clusters of colors** and render an image using just a few colors.



Pixels plotted in **RGB vector space:** 256^3 values.



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Problem with K-means: bad local minima

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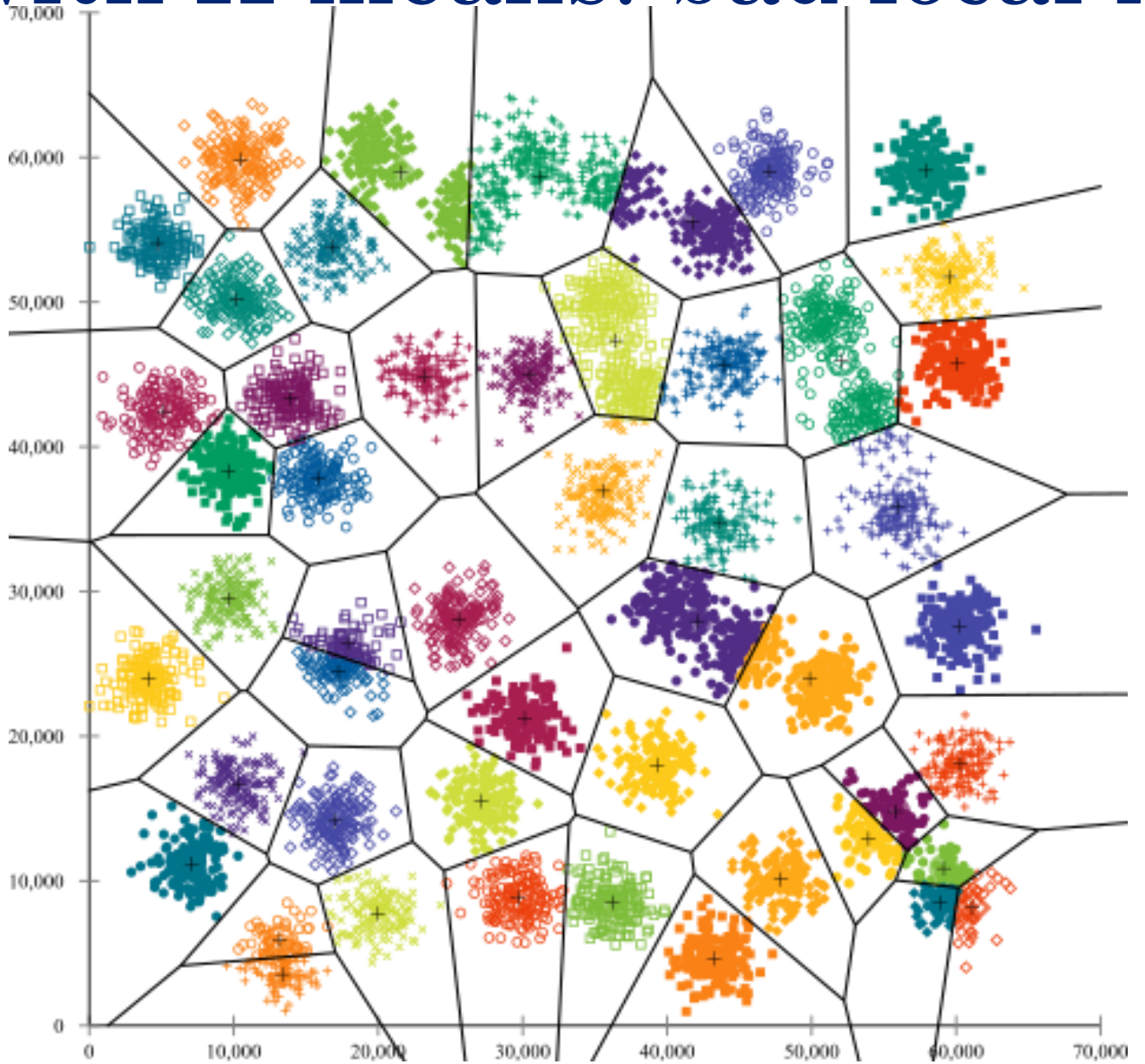

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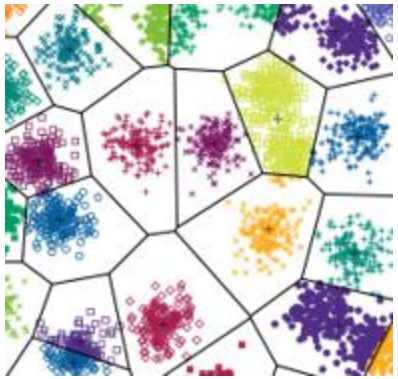
tats.stackexchange.com/questions/133656/how-to-understand-the-drawbacks-of-k-means

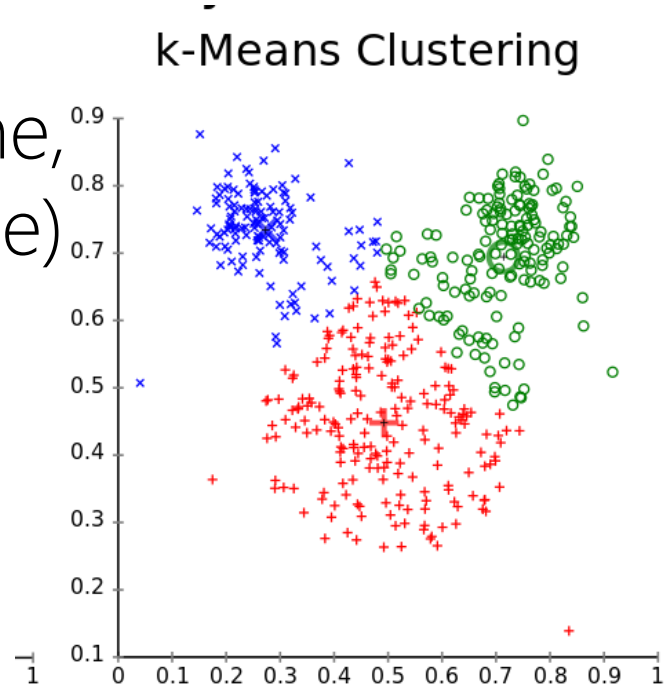
Other weaknesses of k-means clustering

$$\underset{S=C_1 \square \ldots \square C_K,\ \{c_1,\ldots,c_K\}}{\operatorname{argmin}} \square_k \square_{x \square C_k} \|x - c_k\|^2$$



Not probabilistically framed, so hard to understand assumptions and their implications:

- implicitly corresponds to clusters of similar volume, sitting in a "blobby tessellation" (often undesirable)
- also hard to generalize
- each step in the optimization has a "hard" assignment, which feels "brittle"



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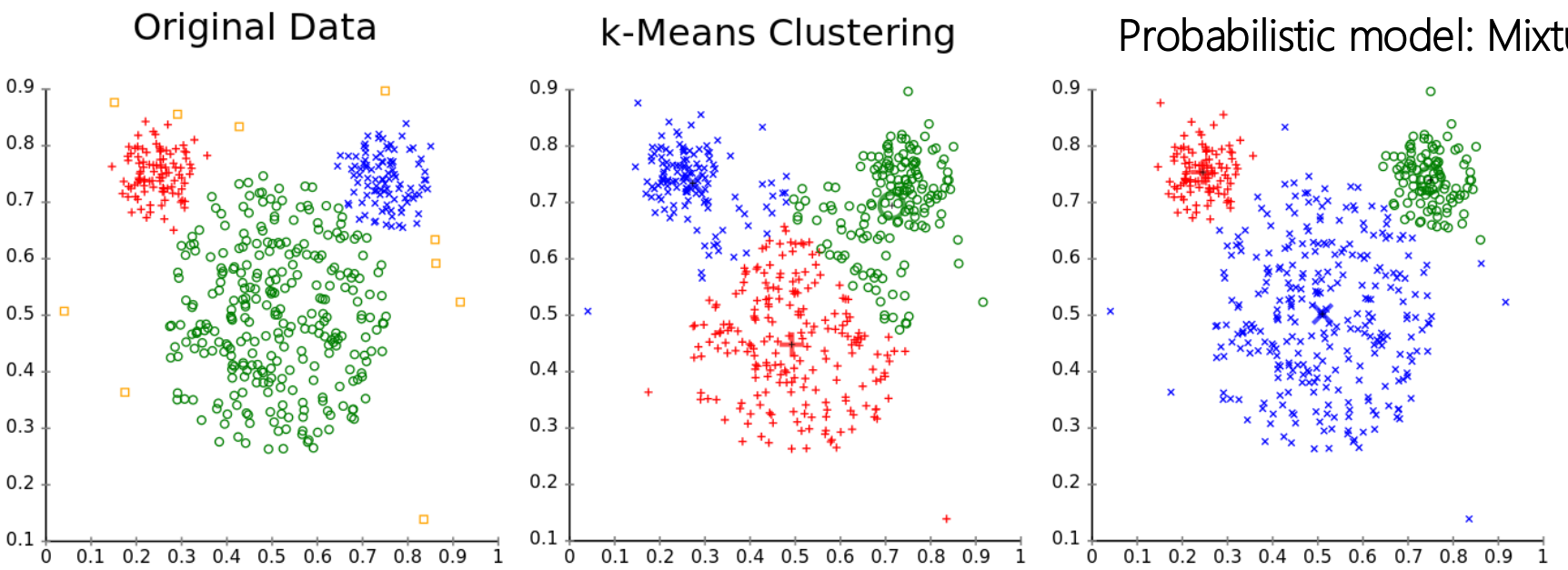
There is a probabilistic version of K-means that fixes these issues! Mixture of Gaussians (next class)



https://en.wikipedia.org/wiki/K-means_clustering

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Roadmap

- Supervised vs Unsupervised Learning
- Clustering
  - K-Means clustering
- Probability (Review)
  - Joint Distributions
  - The IID assumption.

Probability

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Framework for quantifying and manipulating uncertainty.

- **Frequentist view:** about objective, repeatable phenomena. If you flip a fair coin infinitely *many times*, it will land heads 50% of the time.
- **Bayesian view:** represents one's uncertainty, not a physical frequency. *e.g.*, a doctor's belief that a genetic variant is cancer causing in *one particular patient*.

Longstanding debate between these communities. Both are useful.

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The Joint Probability Distribution

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Let X and Y be **discrete random variables** that take values:

$$X \in \{x_1, \dots, x_L\} \quad \text{and} \quad Y \in \{y_1, \dots, y_M\}$$

The joint probability distribution $p(X = x_i, Y = y_j)$ is the probability of the event $X = x_i$ and $Y = y_j$.

Satisfies the **following two properties**:

$p(X = x_i, Y = y_j)$	y_1	y_2
x_1	0.2	0.1
x_2	0.1	0.1
x_3	0.15	0.35

$$\forall x_i, y_j\colon p(X = x_i, Y = y_j) \geq 0 \text{ (Non−negativity)}$$

$$\sum_{i=1}^{L} \sum_{j=1}^{M} p(X = x_i, Y = y_j) = 1 \text{ (Normalization)}$$

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The Joint Probability Distribution

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The joint probability distribution $p(X = x_i, Y = y_j)$ is the probability of the event $X = x_i$ and $Y = y_j$.

The Sum Rule (Marginalization): defines the distribution over a subset of the random variables.

$$p(X = x_i) = \sum_{j=1}^{M} p(X = x_i, Y = y_j)$$

$p(X = x_i, Y = y_j)$	y_1	y_2	$p(X = x_i)$
x_1	0.2	0.1	0.3
x_2	0.1	0.1	0.2
x_3	0.15	0.35	0.5
$p(Y = y_i)$	0.45	0.55	1

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Conditional Probability Distributions

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We define the conditional probability distribution as the probabilities of observing $Y = y$ given **we observed $X = x_i$**

$$p(Y = y_j \mid X = x_i) = \frac{p(X = x_i, Y = y_j)}{p(X = x_i)}$$

Example:

What is $p(Y = y_1 \mid X = x_3)$?

$$\frac{0.15}{0.15 + 0.35} = \frac{0.15}{0.5} = 0.3$$

$$p(Y = y_2 \mid X = x_3) = 1 - 0.3 = 0.7$$

$p(X = x_i, Y = y_j)$	y_1	y_2	$p(X = x_i)$
x_1	0.2	0.1	0.3
x_2	0.1	0.1	0.2
x_3	0.15	0.35	0.5
$p(Y = y_i)$	0.45	0.55	1

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Empirical Probability Distributions

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Consider N trials $\{(X_1, Y_1), \ldots, (X_N, Y_N)\}$ in which we sample (X, Y).

Let n_{ij} be the number (count) of trials where $\left(X = x_i, Y = y_j\right)$, we can construct a discrete **empirical probability distribution:**

$$\hat{p}\left(X = x_i, Y = y_j\right) = \frac{n_{ij}}{N}$$

Frequentist View: In the limit as $N \to \infty$, the empirical probability distribution converges to the true probability distribution:

$$p\left(X = x_i, Y = y_j\right) = \lim_{N \to \infty} \hat{p}\left(X = x_i, Y = y_j\right)$$

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The Product Rule of Probability

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Recall the definition of the conditional:

$$p\big(Y = y_j \mid X = x_i\big) = \frac{p\big(X = x_i, Y = y_j\big)}{p(X = x_i)}$$

Multiply both sides by $p(X = x_i)$ you obtain the **product rule:**

$$p\big(X = x_i, Y = y_j\big) \;=\; p\big(Y = y_j \mid X = x_i\big)\, p(X = x_i)$$

The product rule is like a **chain rule** of probability – it allows us to **factorize** joint probability distributions into conditionals:

$$p(X, Y, Z) = p(Z \mid X, Y)\, p(X|Y)\, p(Y)$$

- *Notation:* we often drop x_i, y_j, and z_k for convenience.

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Independent Random Variables

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Two variables X and Y are independent if the joint probability factorizes:

$$X \perp Y \implies \boldsymbol{p}(\boldsymbol{X}, \boldsymbol{Y}) = \boldsymbol{p}(\boldsymbol{X})\, \boldsymbol{p}(\boldsymbol{Y})$$

Using the definition of conditionals when $X \perp Y$:

$$\implies \boldsymbol{p}(\boldsymbol{Y}|\boldsymbol{X}) = \frac{p(Y, X)}{p(X)} = \frac{p(Y)p(X)}{p(X)} = \boldsymbol{p}(\boldsymbol{Y})$$

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Independent & Identically Distributed Data

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If data $x_1, x_2, \dots, x_N$ are **Independent and Identically Distributed** (IID) then their joint probability factorizes:

$$p(x_1, x_2, \dots x_N) = \prod_{n=1}^{N} p(X = x_n)$$

- We will often assume IID data (though it is often not true in practice). Did we make this assumption for K-means?
- Analogous definition for joint and conditionals (any prob. distrib):
  - $p(x_1, y_1, \dots x_N, y_N) = \prod_{n=1}^{N} p(X = x_n, Y = y_n)$
  - $p(y_1, \dots, y_N | x_1, \dots x_N) = \prod_{n=1}^{N} p(Y = y_n | X = x_n)$

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Lecture 4

K-Means and Probability

Reference Book Chapters:

- **Clustering:** Chapter 15.1 (k-means)
- **Probability:** Chapter 2.[1-2]

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Roadmap

EXTRAS:

Probabilities in action

- Wake Word Example (Bayes' Rule)

Example: Wake Words

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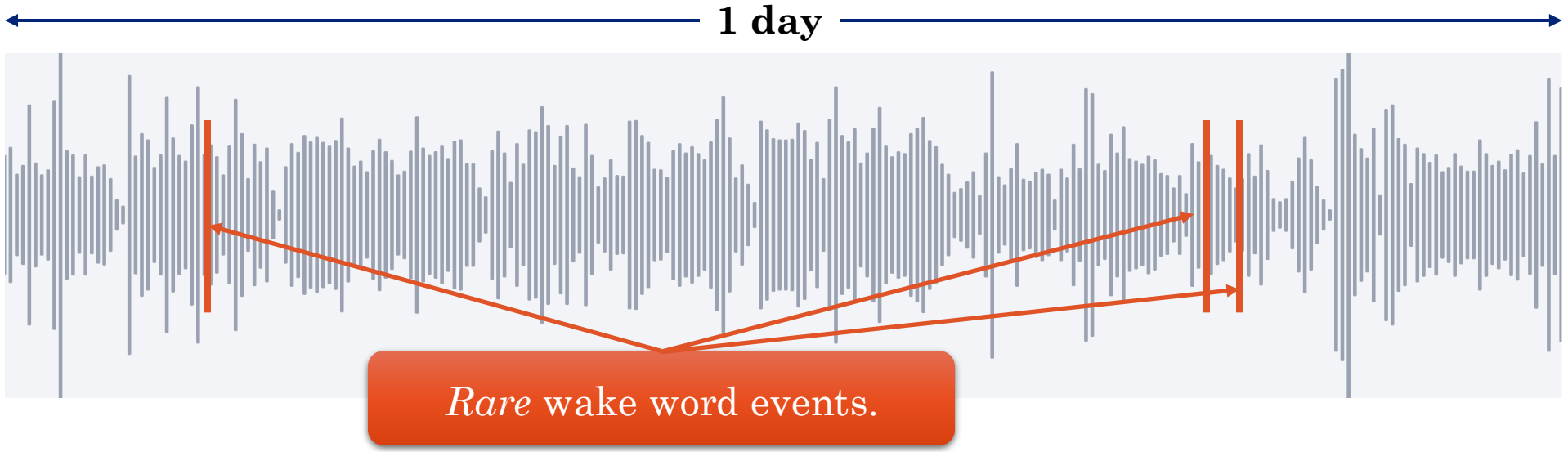
A **wake word** is a verbal cue that triggers voice assistants to start actively listening.

Example: *"**Alexa**, set an alarm for 6:00AM"*

Most voice assistants **continuously** run a wake word detector model on every sound they hear.



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Example: Wake Words

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How good does the wake word detector model need to be?

There are two relevant random variables:

1. Truth: was a wake word said: $X \in \{0, 1\}$ (0: no, 1: yes)
2. Empirics: did we detect a wake word: $Y \in \{0, 1\}$ (0: negative, 1: positive)
3. What concepts do these 4 probabilities represent?

$$p(X = 1 \mid Y = 1) \qquad p(X = 0 \mid Y = 1)$$
$$p(X = 1 \mid Y = 0) \qquad p(X = 0 \mid Y = 0)$$

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Analyzing the Wake Word Detector

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Let's assume wake words are rare occurring less than 0.01% of segments:

$$p(X = 1) = 0.0001 \quad \text{and} \quad p(X = 0) = 1 - p(X = 1) = 0.9999$$

Suppose the wake word model **detects 99% of the wake words**:

$$p(Y = 1 \mid X = 1) = 0.99$$

And **falsely detects (false positive)** a wake word just 0.1% of the time:

$$p(Y = 1 \mid X = 0) = 0.001$$

If the **detector is positive**, what is the chance that **it was a wake word**?

$$p(X = 1 \mid Y = 1) = \frac{p(Y = 1, X = 1)}{p(Y = 1)} = \frac{p(Y = 1 | X = 1)\, p(X = 1)}{p(Y = 1)}$$

Bayes' Theorem

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Bayes' Theorem

1543476

Named after Thomas Bayes who was an 18th century statistician, philosopher, and minister.

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)} = \frac{p(B|A)p(A)}{\sum_A p(B|A)p(A)}$$

- Derived from product rule: $p(A,B) = p(A|B)p(B) = p(B|A)p(A)$

Useful in Bayesian statistics (but also in manipulating probabilities):

- $p(A)$ is the **prior belief** about A
- $p(B|A)$ is the **likelihood** of an observation B given those beliefs
- $p(A|B)$ is our **posterior belief** about A given our observation B

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Analyzing the Wake Word Detector

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We assumed rare wake words: $p(X = 1) = 0.0001$

A **true positive rate (recall)** of 99%: $p(Y = 1 \mid X = 1) = 0.99$

And a **false positive rate of** 0.1%: $p(Y = 1 \mid X = 0) = 0.001$

If the **detector is positive**, what is the chance that **it was a wake word**?

$$p(X = 1 \mid Y = 1) = \frac{p(Y = 1|X = 1)\, p(X = 1)}{p(Y = 1|X = 1)\, p(X = 1) + p(Y = 1|X = 0)\, p(X = 0)}$$

$$p(X = 1 \mid Y = 1) = \frac{0.99 * 0.0001}{0.99 * 0.0001 + 0.001 * 0.9999} = 0.09 = 9\%$$

Is that good (enough)?

How would we improve it?

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Bayesian Updates: Wake Word Detector

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Prior: we assumed rare wake words $p(X = 1) = 0.0001$

After observing the test $Y = 1$ we **updated our belief** about the presence of the wake word using Bayes Rule:

$$p(X = 1 \mid Y = 1) = \frac{p(Y = 1, X = 1)}{p(Y = 1)} = \frac{p(Y = 1 | X = 1)\, p(X = 1)}{p(Y = 1)} = 0.09$$

Posterior: the updated belief $p(X = 1 \mid Y = 1)$ after the observation.

Bayesian Updating: the process of starting from a **prior belief** and then using **observations** to compute **posterior distribution** (updated belief) is central to **Bayesian machine learning**.

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